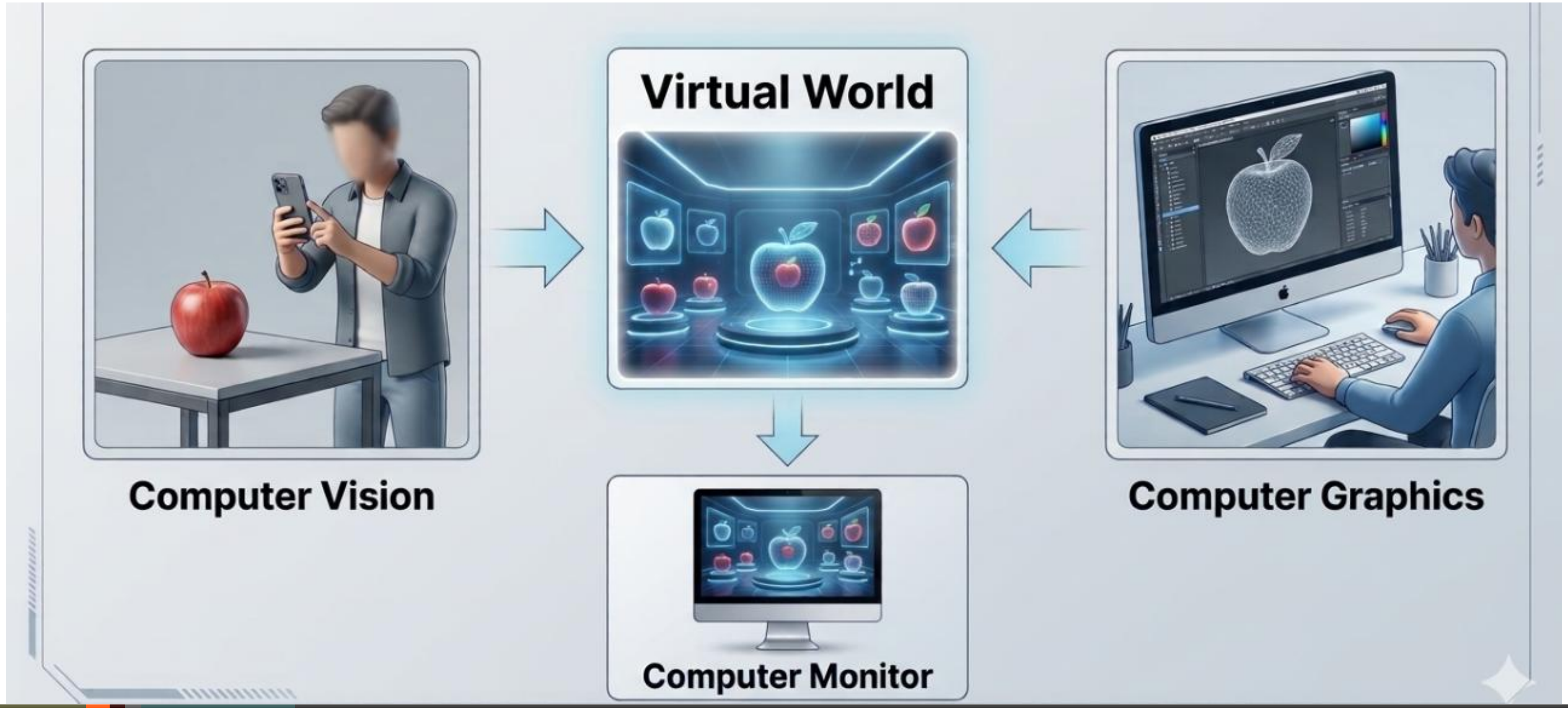
13. Beyond Polygons



Beyond Polygons

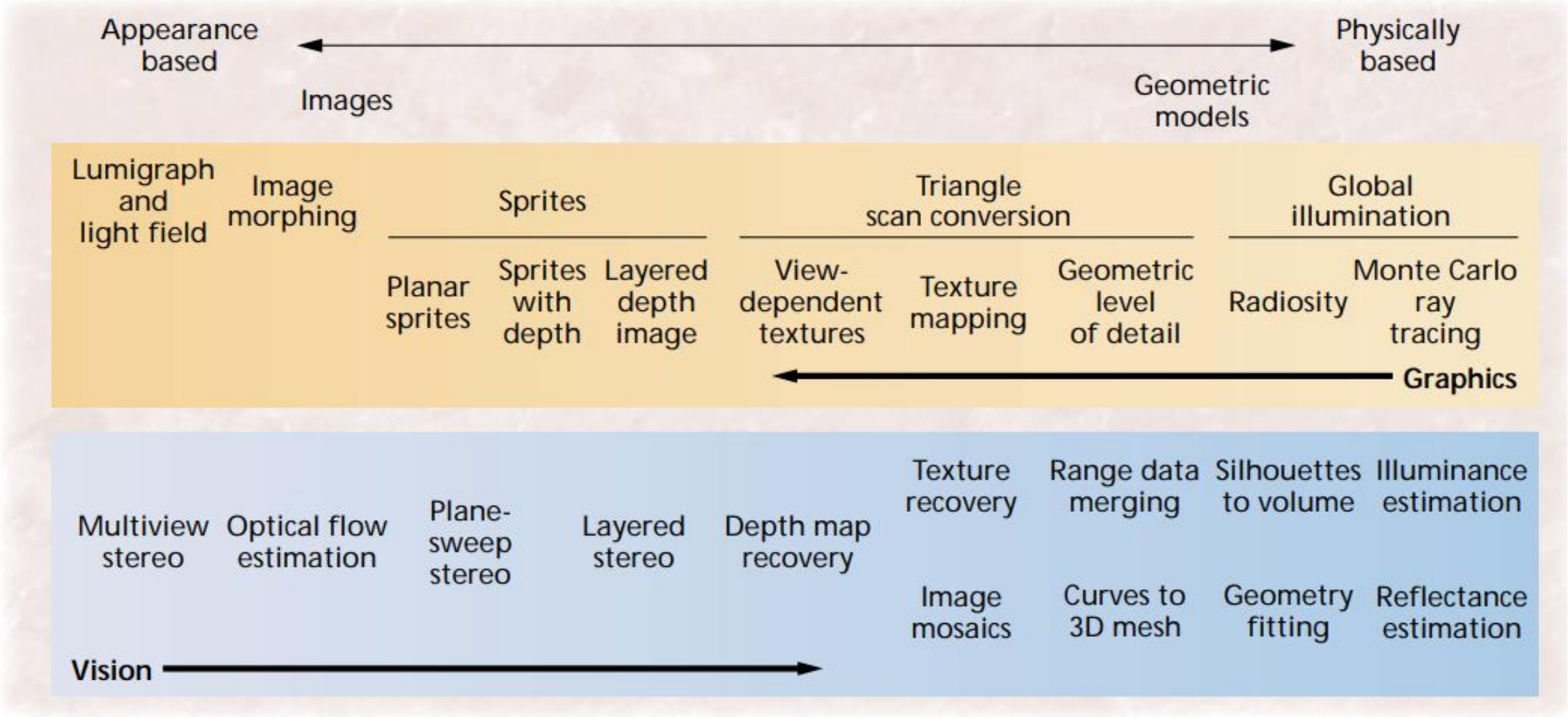
Chapter Overview



자료 출처 : Google Nano Banana 2

Beyond Polygons

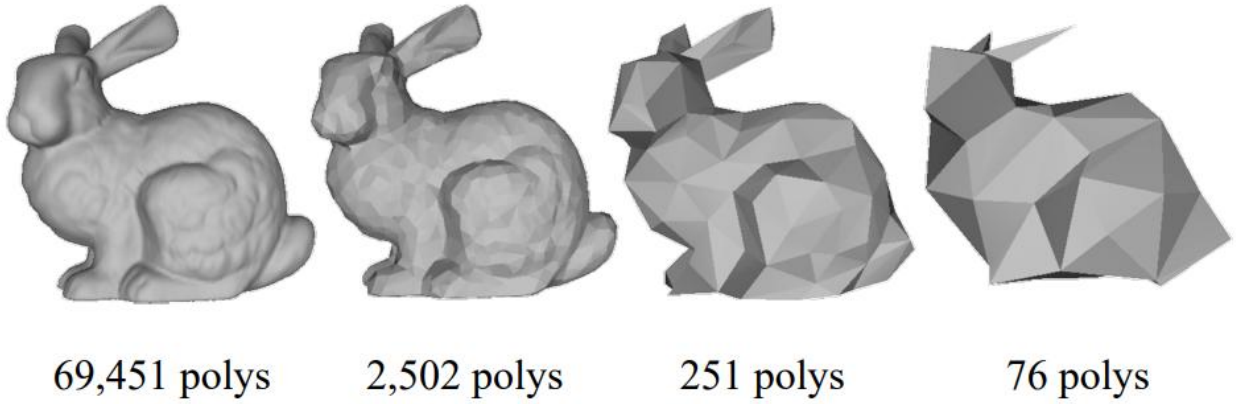
The Rendering Spectrum

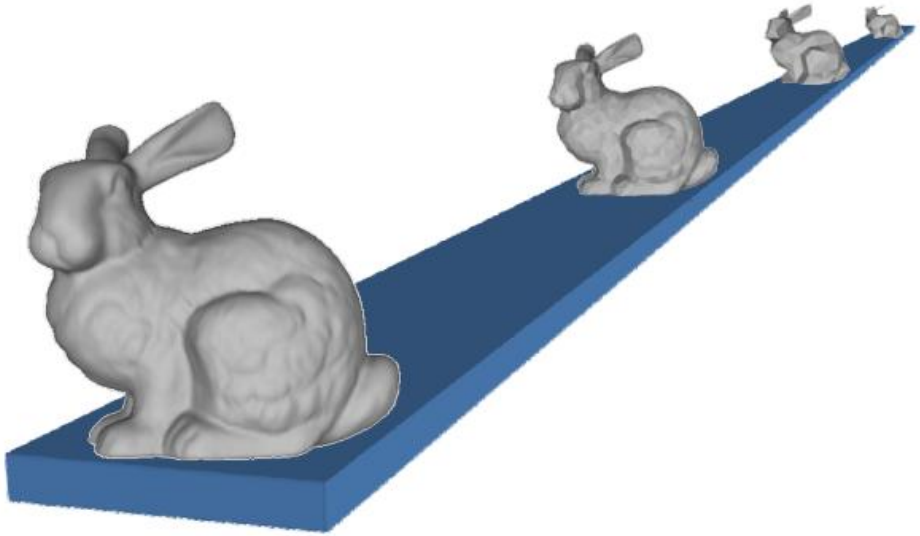


자료 출처 : Lengyel, Jerome, "The Convergence of Graphics and Vision," Computer, vol. 31, no. 7, pp. 46-53, July 1998

LOD

- Level of Detail
  - 객체가 가까울수록 상세하게, 멀수록 단순하게
  - 렌더링을 더 빠르게 하는 것이 핵심





LOD (Level of Detail)

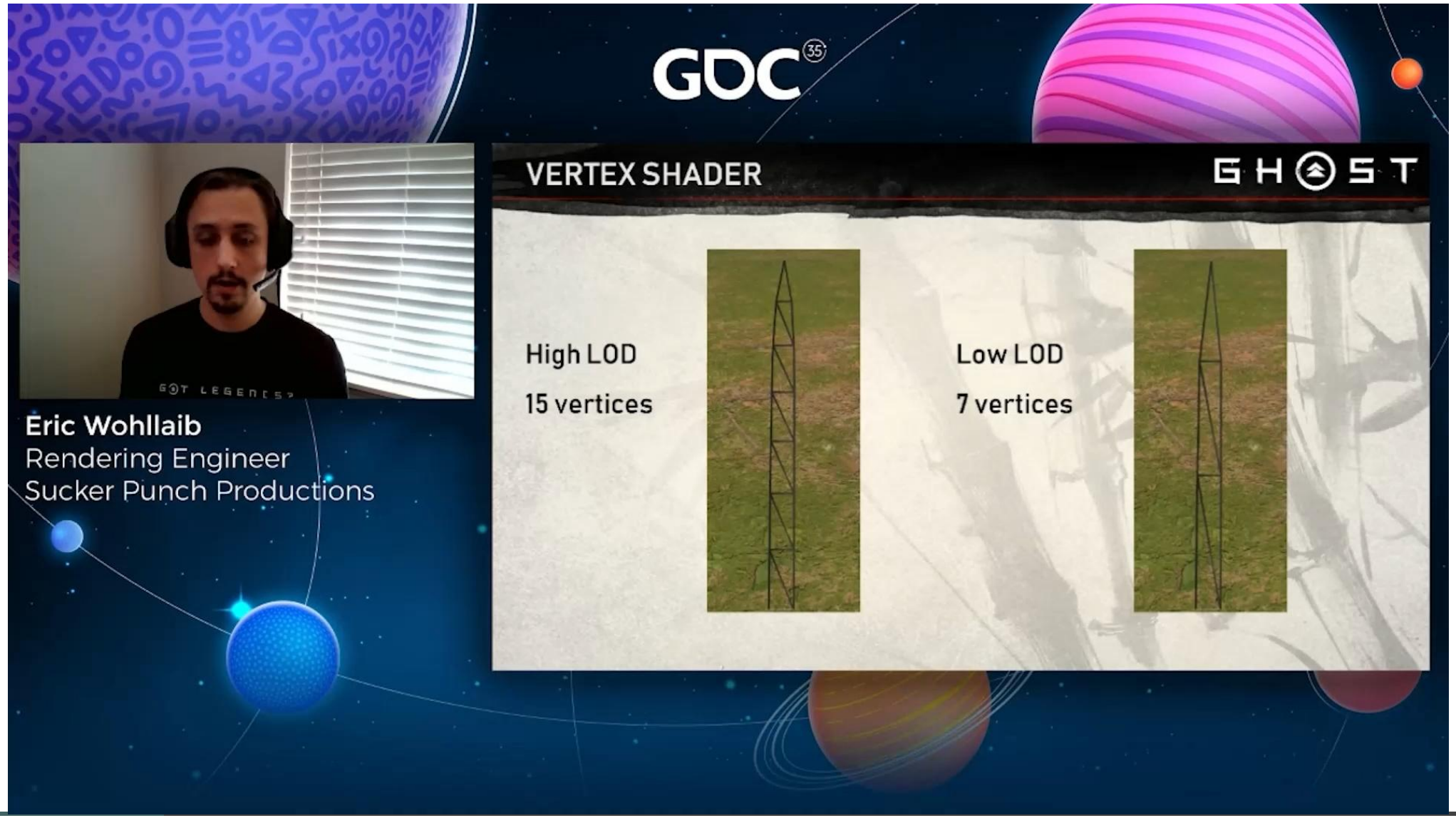
Crimson Desert LOD Problems



자료 출처 : https://youtu.be/wUIGZBRdcXY?si=rE6N0TGOFvipGLOS (Crimson Desert LOD Problems)

LOD (Level of Detail)

Procedural Grass in ‘Ghost of Tsushima’



자료 출처 : https://youtu.be/Ibe1JBF5i5Y?si=qNRuAu-MJUNJJyQt (2021 GDC, Advanced Graphics Summit talk)

The Rendering Spectrum

Fixed-View Effects (1/4)

1 정적 장면 렌더링 (목장과 울타리)

카메라가 고정된 상태에서 정적 장면(목장과 울타리)만 렌더링한다.



이 시점의 Color Buffer와 Z Buffer를 저장해 둔다.

버퍼 저장

Color Buffer (색상 버퍼)



Z Buffer (깊이 버퍼)



자료 출처 : ChatGPT Free

Fixed-View Effects (2/4)

2 저장된 버퍼로 초기화

각 프레임마다 저장된 Color Buffer와 Z Buffer를 화면 버퍼로 복원한다.

초기화된 Color Buffer



초기화된 Z Buffer



이제 화면(Color, Z-buffer)은 정적 장면으로 초기화되어 있다.
이후에는 변경되는 객체(말)만 렌더링하면 된다.

자료 출처 : ChatGPT Free

Fixed-View Effects (3/4)

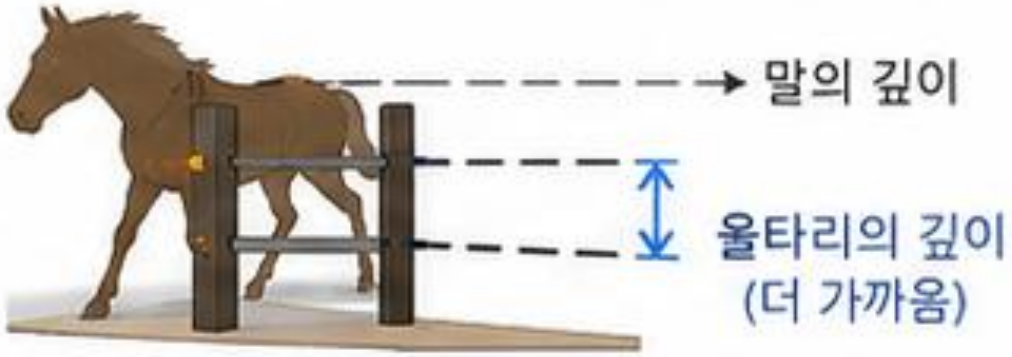
3 동적 객체만 렌더링 (말)

말(동적 객체)만 렌더링한다. Z-buffer 테스트에 의해 가려질 부분은 자동으로 제거된다.

렌더링할 말



Z Buffer 테스트 예시 (단면도)



울타리가 말보다 더 가까운 픽셀은
Z-buffer 값이 더 작으므로
말의 픽셀이 그려지지 않는다.

말의 깊이 (Z 값)



말의 색상 (Color)



자료 출처 : ChatGPT Free

Fixed-View Effects (4/4)



- 카메라가 고정된 상황에서는 복잡한 정적 장면을 매 프레임 다시 렌더링할 필요 없음
  - 저장된 Color/Z 버퍼를 복사하는 것만으로 초기화가 매우 빠름
  - 동적 객체만 렌더링하므로 성능이 크게 향상
- CAD 및 Visualization 시스템에서 자주 사용
- 시간이 지남에 따라 더 나은 이미지를 생성할 수 있음
  - 골든 스레드(Golden thread)
  - 적응 개선(Adaptive refinement)
  - 점진적 개선(Progressive refinement)

자료 출처 : ChatGPT Free

Beyond Polygons

Skyboxes

1. 환경 맵 (Environment Map) 예시

파노라마(등거리) 환경 맵



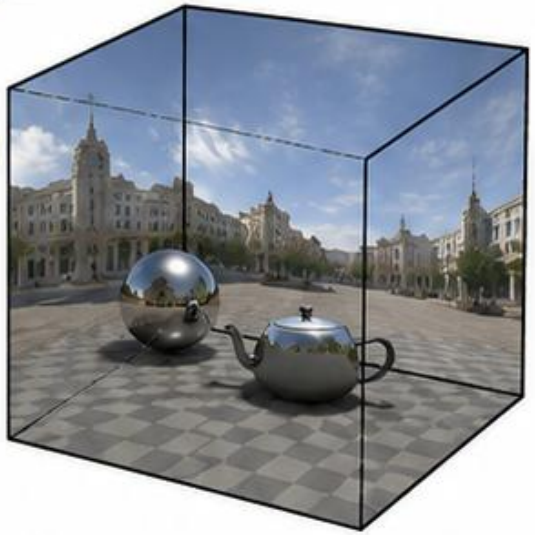
큐브 맵 (6개 면)



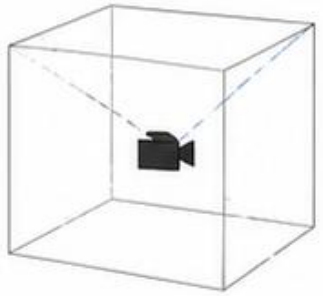
2. 스카이박스 구조

스카이박스 메시
(내부가 보이도록
면의 법선이 안쪽을 향함)

환경 맵을 박스의
안쪽 면에 적용



카메라가 박스 내부에 위치



스카이박스는 매우 크게 만들어
모든 다른 객체를 감싸도록 한다.

- 스카이박스가 보기 좋게 보이려면 큐브 맵 텍스처 해상도가 충분해야 함
  - 텍스처 해상도 $= \frac{\text{화면 해상도}}{\tan(fov\ /\ 2)}$
  - 큐브 맵의 한 면의 텍스처가 90°의 시야를 커버해야 한다는 관찰에서 파생
- 스카이박스는 아무것도 차단하지 않기 때문에 Z버퍼에 쓰일 필요가 없음

Beyond Polygons

Skyboxes in Video Games

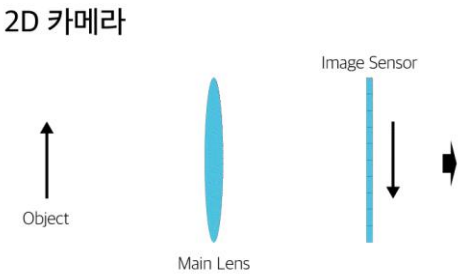


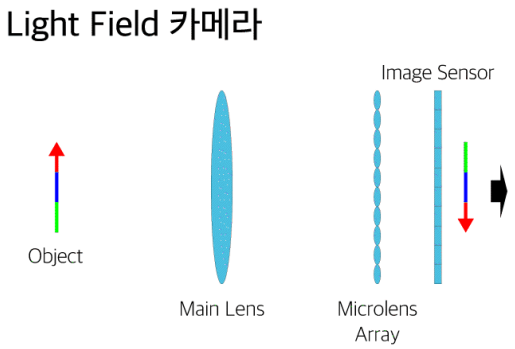
자료 출처 : youtu.be/lb7xwB-JvWs?si=WSmlDgX055RFCXXP (Blurbs, Skyboxes in Video Games), youtu.be/LOBJGzP4DU8?si=n66wCJCJs_I49lO6 (IKIGAI, The Beauty of Elden Ring (4K Maximum Graphics)

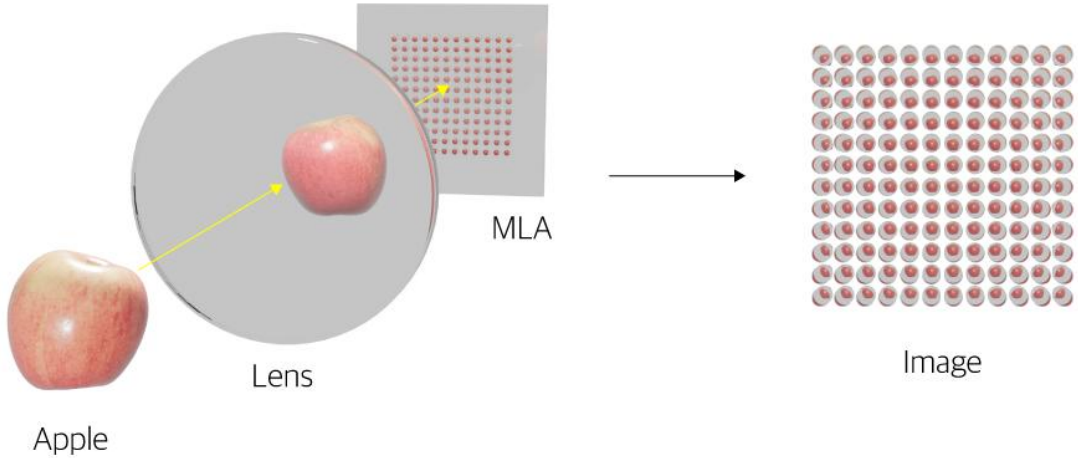
Beyond Polygons

Light Field Rendering

- "Computational Photography: Mastering New Techniques for Lenses, Lighting, and Sensors" [Ramesh Raskar and Jack Tumblin / A. K. Peters, Ltd, 2009]
- "Light Field Rendering" [Marc Levoy and Pat Hanrahan / SIGGRAPH, 1996]







자료 출처 : https://www.linkedin.com/pulse/기존-3d-비전의-한계-라이트필드는-어떻게-넘었나-kais-ijy3c

Sprites & Layers

- Sprite (스프라이트)
  - 마우스 커서와 같이(처럼) 화면에서 움직이는 이미지
  - 항상 관측자를 향하는 폴리곤에 적용된 이미지 텍스처로 렌더링하는 것

    ⇒ 독립적으로 움직일 수 있는 이미지





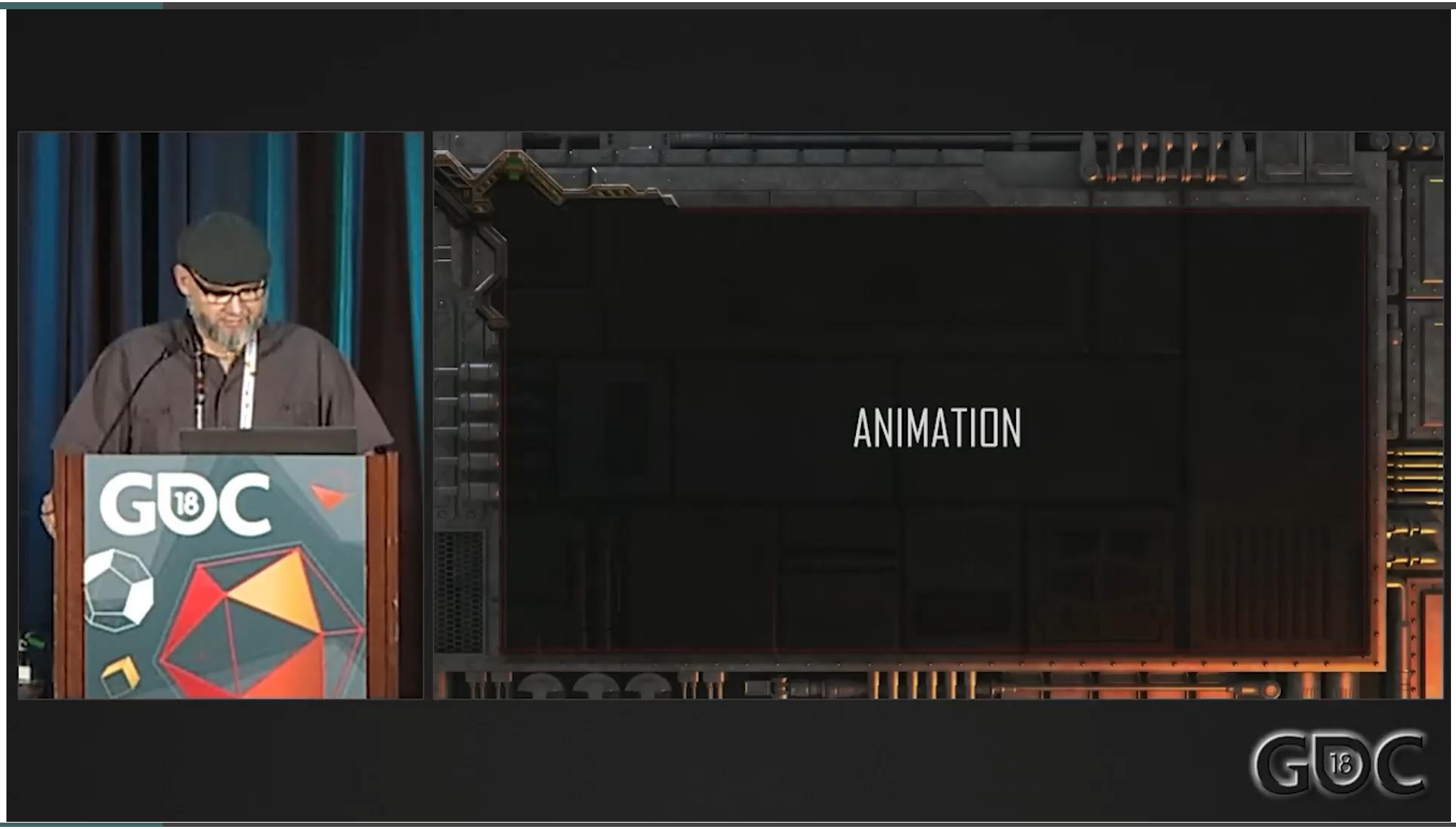
자료 출처 : https://www.spriters-resource.com/sega_cd/sonicthehedgehogcd/asset/10831/

Chicken Crossing



자료 출처 : Chicken crossing – By Andrew Glassner, for MSTalisman (https://youtu.be/yFmG-gkVnHY?si=Ggh4vXDGbuGCfNXS)

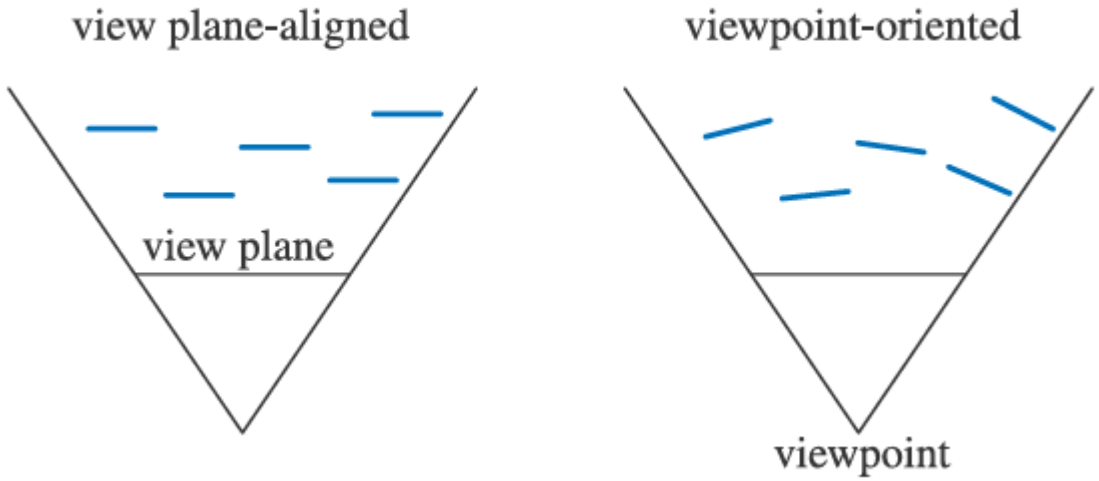
StartCraft Remastered - Animation

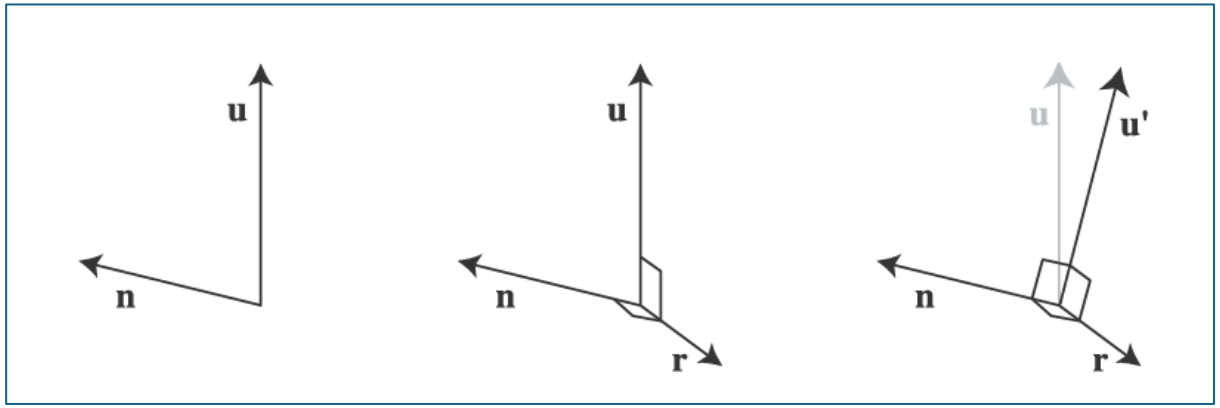


자료 출처 : 2018 GDC - Game Art Remastered: From Crash, to Kerrigan, to Corley Motors (https://youtu.be/6NSEzm0qy1s?si=zm31lkBG4CFbFMwj)

Billboard

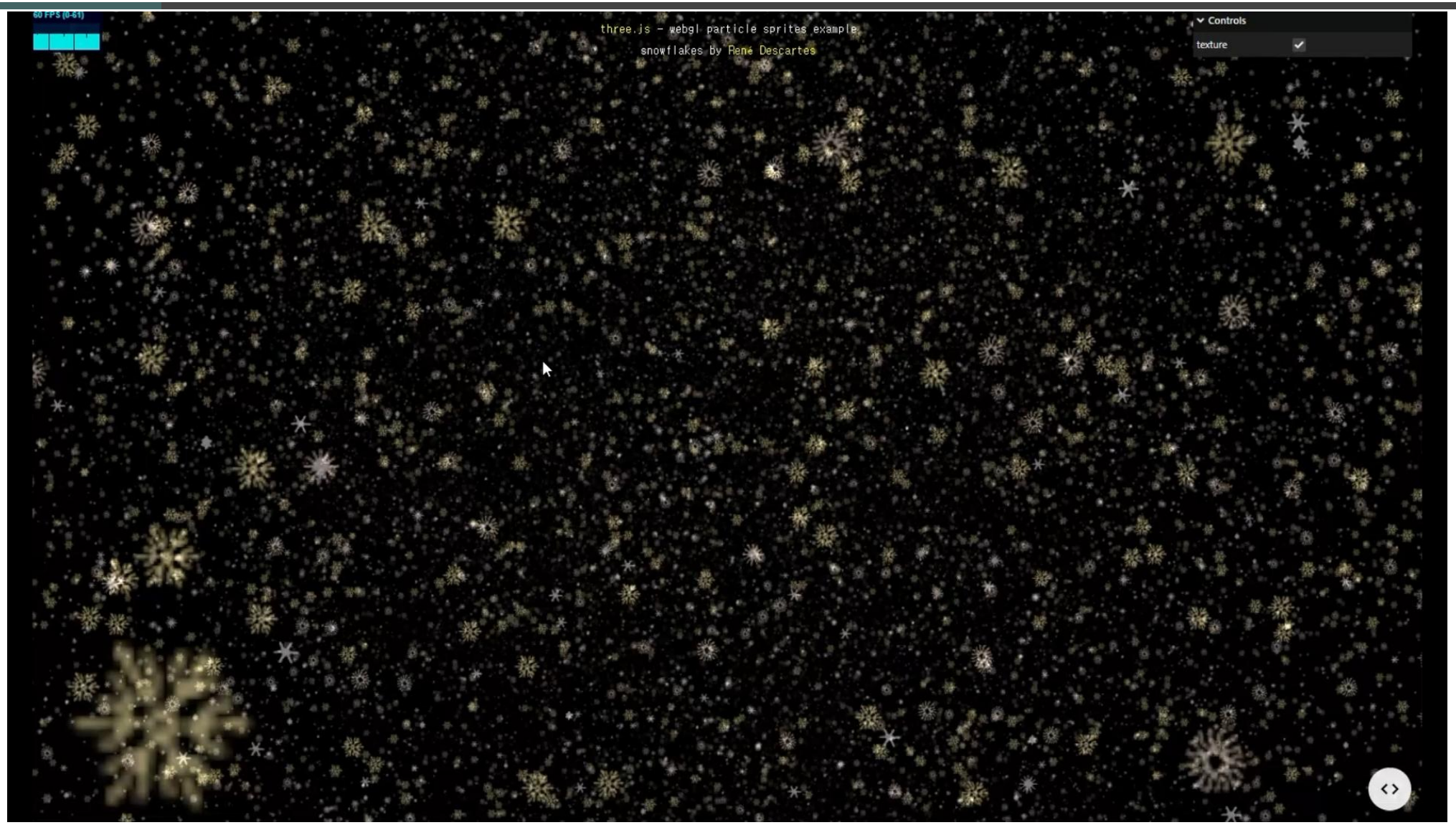
- 빌보딩(Billboarding)
  - 뷰 방향을 기준으로 텍스처링된 직사각형의 방향을 지정하는 것
  - 여기서의 직사각형을 빌보드(Billboard)
    - 뷰가 변경되면 직사각형의 방향도 수정





Billboard

three.js example



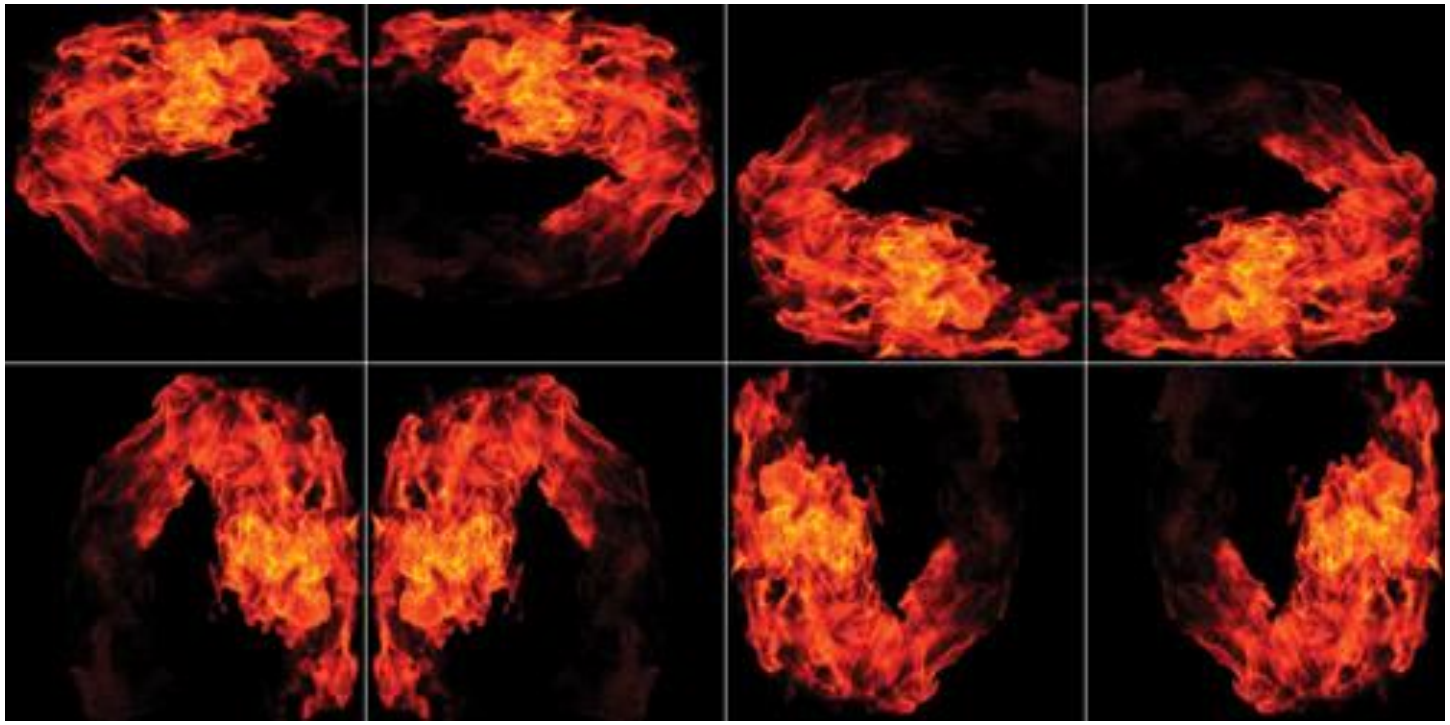
자료 출처 : https://threejs.org/examples (points/sprites , buffergeometry/instancing/billboards , points/dynamic)

Billboard in Video Games

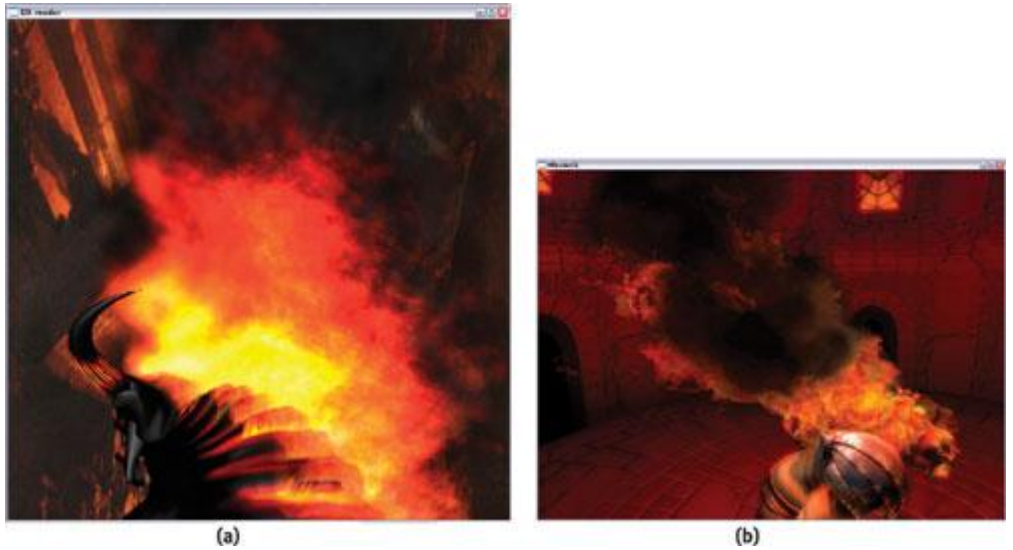


자료 출처 : Masahiro Sakurai on Creating Games – Billboards [Effects] (https://youtu.be/_o29OOOarPY?si=26FC_PuYfdQjwh_G)

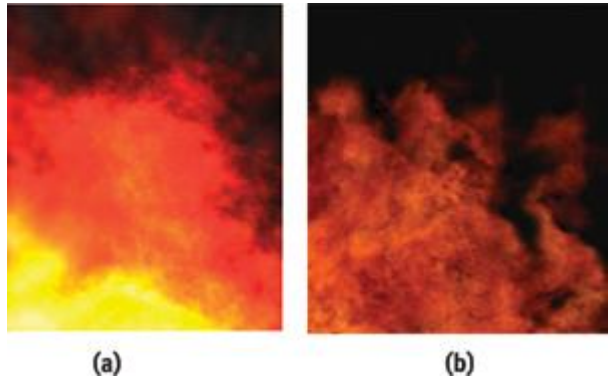
Fire in the 'Vulcan' Demo by NVIDIA (1/2)



Variations Produced by Custom Texture Coordinates



Mixing Fire and Smoke



Controlling Color

자료 출처 : https://developer.nvidia.com/gpugems/gpugems/part-i-natural-effects/chapter-6-fire-vulcan-demo

Billboard

Fire in the ‘Vulcan’ Demo by NVIDIA (2/2)



자료 출처 : nVidia Vulcan Demo (HD) (https://youtu.be/tuj4bGc5ZTk?si=EFMw8rvm6BPQbNgK)

Billboard

Cutout Texture



자료 출처 : Advanced in Real-Time Rendering in Games course – SIGGRAPH 2012 (https://advances.realtimerendering.com/s2012/index.html)

Particle trimming

Particle trimming

- GPUs increasingly more powerful
  - ALU – Through the roof
  - TEX – Pretty decent increase
  - BW – Kind of sluggish
  - ROP – Glacial speed
- ROP bound?
  1. Draw fewer pixels
  2. ???
  3. Goto 1

	9700 Pro (2002)	HD 2900XT (2007)	HD 7970 (2012)	10 years peedup
ALU	33.8 GF/s	475 GF/s	3789 GF/s	112x
TEX	2.6 GT/s	11.9 GT/s	118.4 GT/s	46x
BW	19.84 GB/s	105.6 GB/s	288 GB/s	15x
ROP	2.6 GP/s	11.9 GP/s	29.6 GP/s	11x

Source: Wikipedia [1]

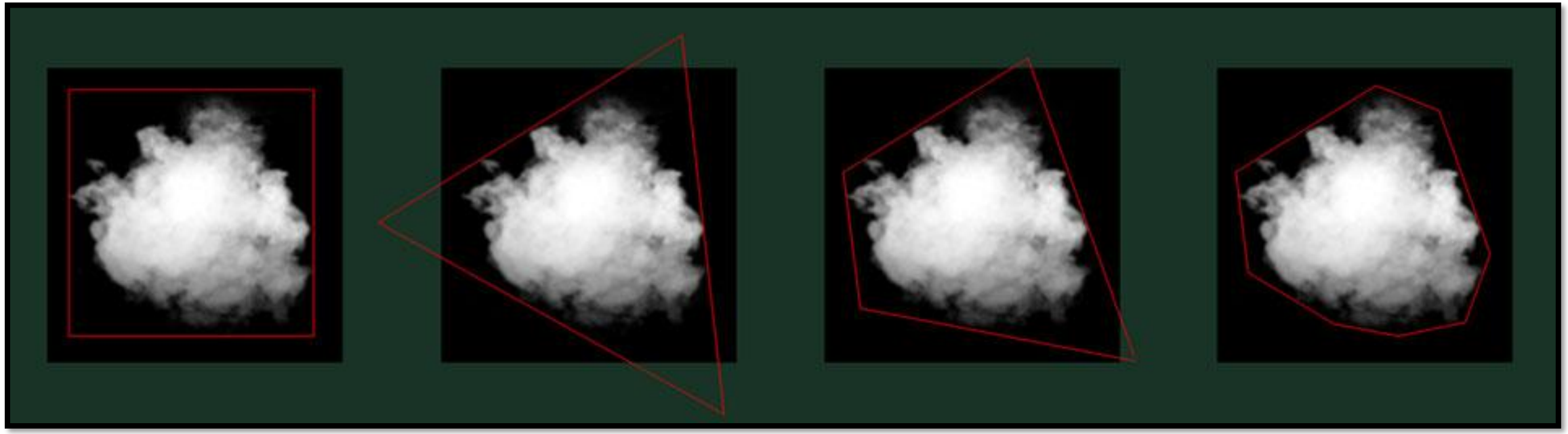
자료 출처 : Advanced in Real-Time Rendering in Games course – SIGGRAPH 2012 (https://advances.realtimerendering.com/s2012/Avalanche/Persson_GraphicsGemsForGames.pptx)

Particle trimming

- Typical ROP bound cases
  - Particles
  - Clouds
  - Billboards
  - GUI elements
- Solutions
  - Render to low-res render target
  - Abuse MSAA
- Our solution
  - Trim particle polygon to reduce waste

Particle trimming

- Common with large alpha=0 areas
  - Wasted fillrate
  - Adjust particle geometry to minimize waste
  - Automated tool [2]

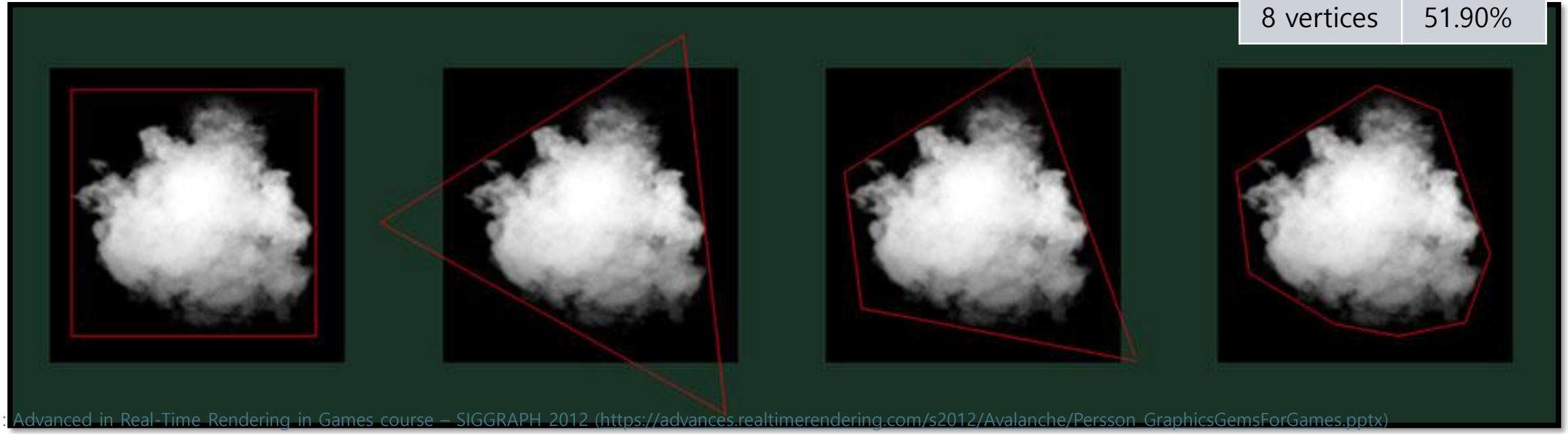


자료 출처 : Advanced in Real-Time Rendering in Games course – SIGGRAPH 2012 (https://advances.realtimerendering.com/s2012/Avalanche/Persson_GraphicsGemsForGames.pptx)

Particle trimming

- Huge fillrate savings
  - More vertices ⇒ Bigger saving
  - Diminishing returns
  - Just Cause 2 used 4 for clouds, 8 for particle effects

Original	100%
Tight rect	69.23%
3 vertices	70.66%
4 vertices	60.16%
5 vertices	55.60%
6 vertices	53.94%
7 vertices	52.31%
8 vertices	51.90%



자료 출처 : Advanced in Real-Time Rendering in Games course – SIGGRAPH 2012 (https://advances.realtimerendering.com/s2012/Avalanche/Persson_GraphicsGemsForGames.pptx)

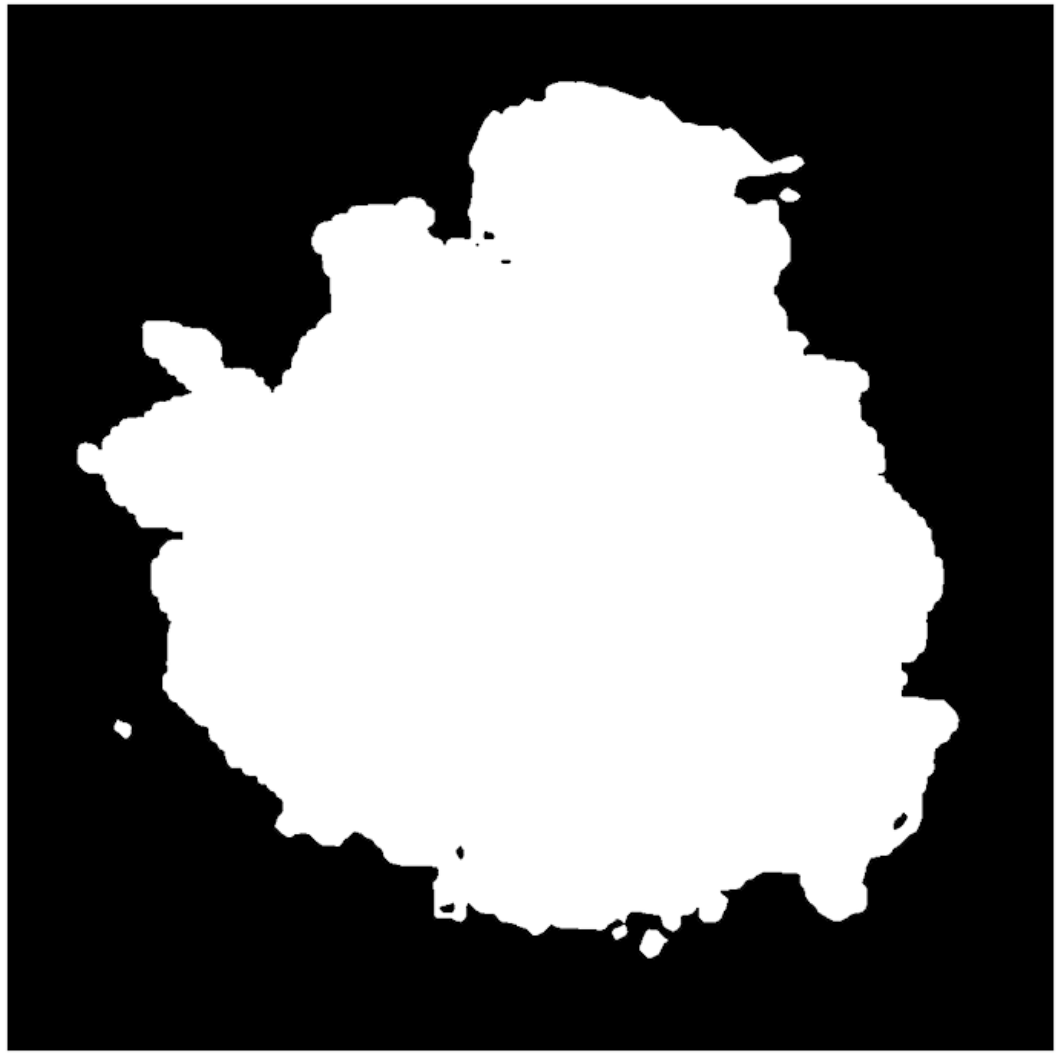
Particle trimming



Original texture

자료 출처 : Advanced in Real-Time Rendering in Games course – SIGGRAPH 2012 (https://advances.realtimerendering.com/s2012/Avalanche/Persson_GraphicsGemsForGames.pptx)

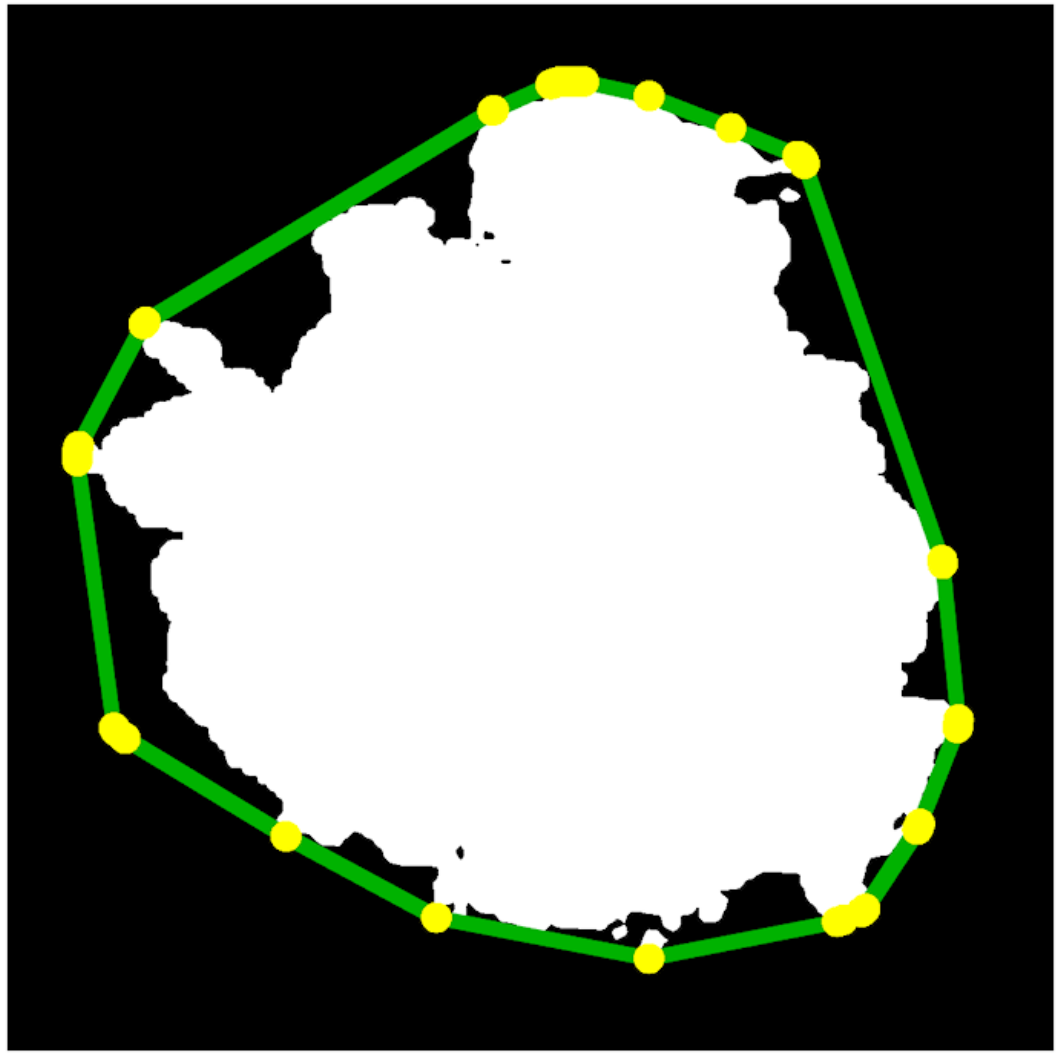
Particle trimming



Thresholded texture

자료 출처 : Advanced in Real-Time Rendering in Games course – SIGGRAPH 2012 (https://advances.realtimerendering.com/s2012/Avalanche/Persson_GraphicsGemsForGames.pptx)

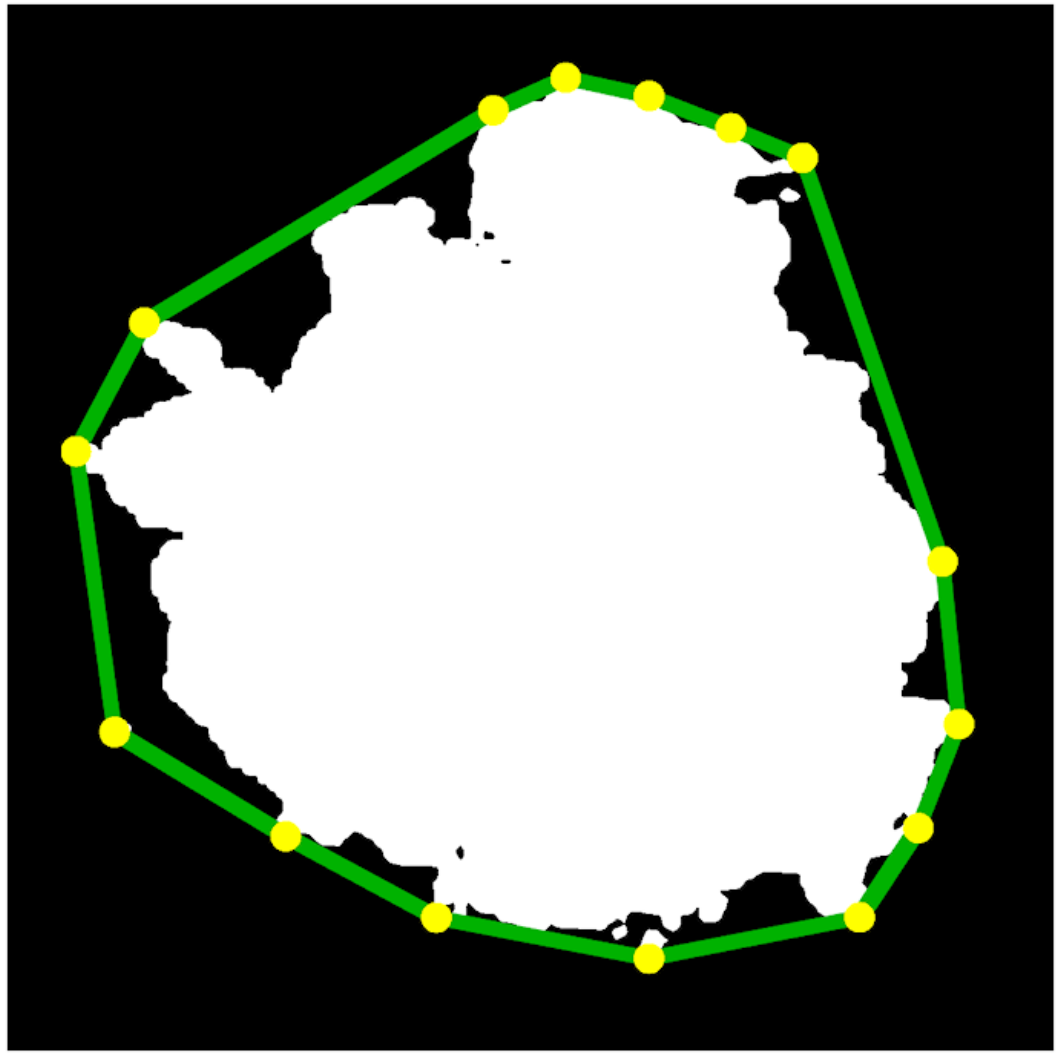
Particle trimming



Convex hull

자료 출처 : Advanced in Real-Time Rendering in Games course – SIGGRAPH 2012 (https://advances.realtimerendering.com/s2012/Avalanche/Persson_GraphicsGemsForGames.pptx)

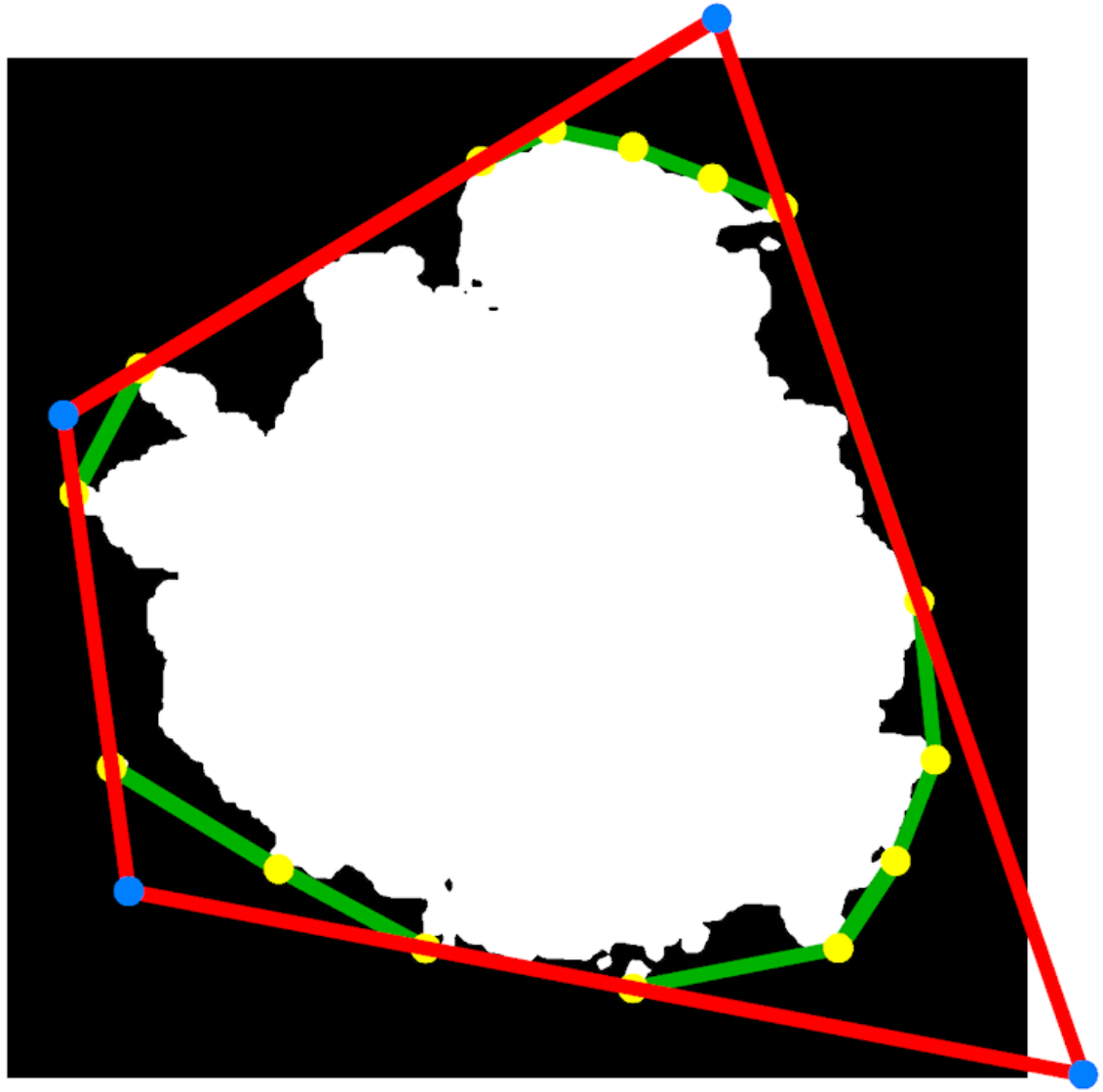
Particle trimming



Reduced convex hull

자료 출처 : Advanced in Real-Time Rendering in Games course – SIGGRAPH 2012 (https://advances.realtimerendering.com/s2012/Avalanche/Persson_GraphicsGemsForGames.pptx)

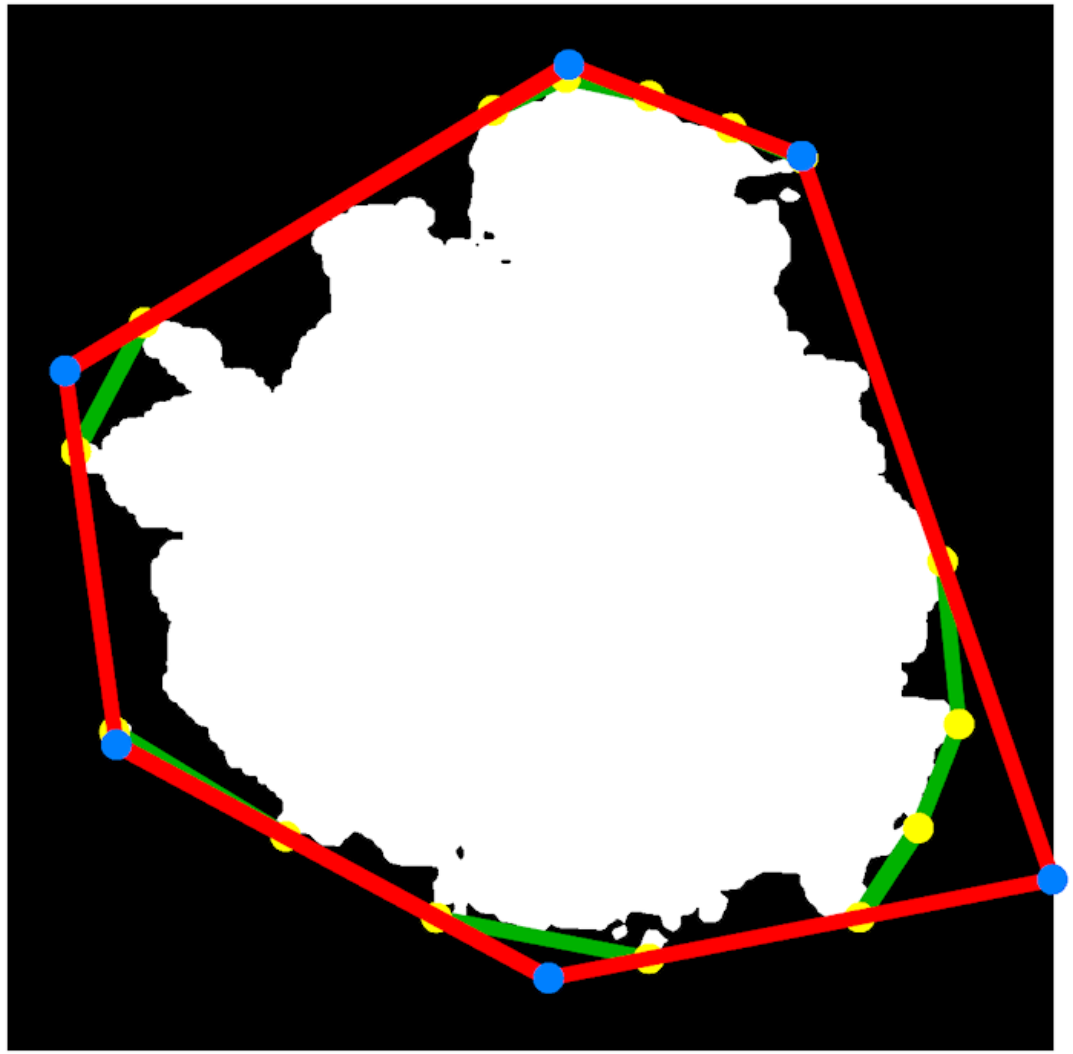
Particle trimming



Final 4 vertex polygon (60.16%)

자료 출처 : Advanced in Real-Time Rendering in Games course – SIGGRAPH 2012 (https://advances.realtimerendering.com/s2012/Avalanche/Persson_GraphicsGemsForGames.pptx)

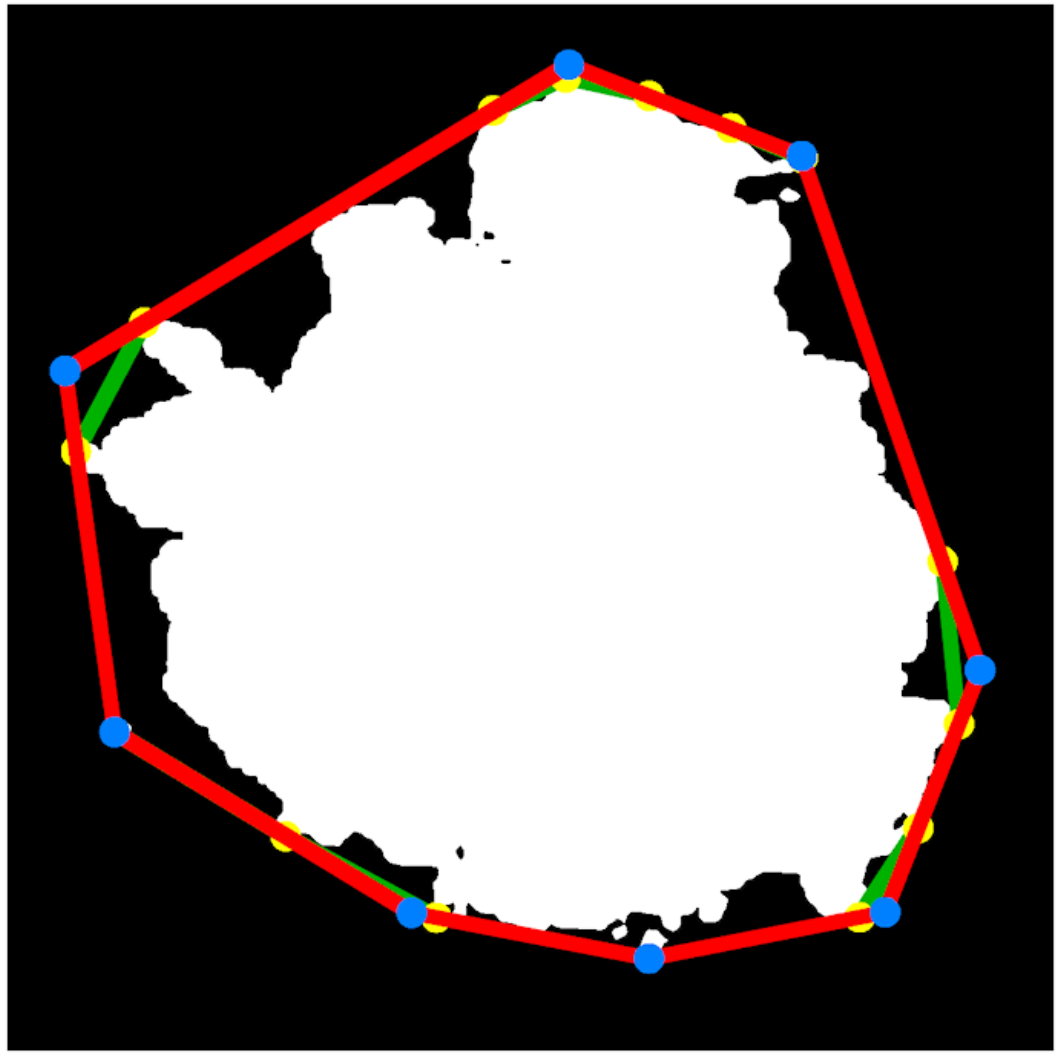
Particle trimming



Final 6 vertex polygon (53.94%)

자료 출처 : Advanced in Real-Time Rendering in Games course – SIGGRAPH 2012 (https://advances.realtimerendering.com/s2012/Avalanche/Persson_GraphicsGemsForGames.pptx)

Particle trimming

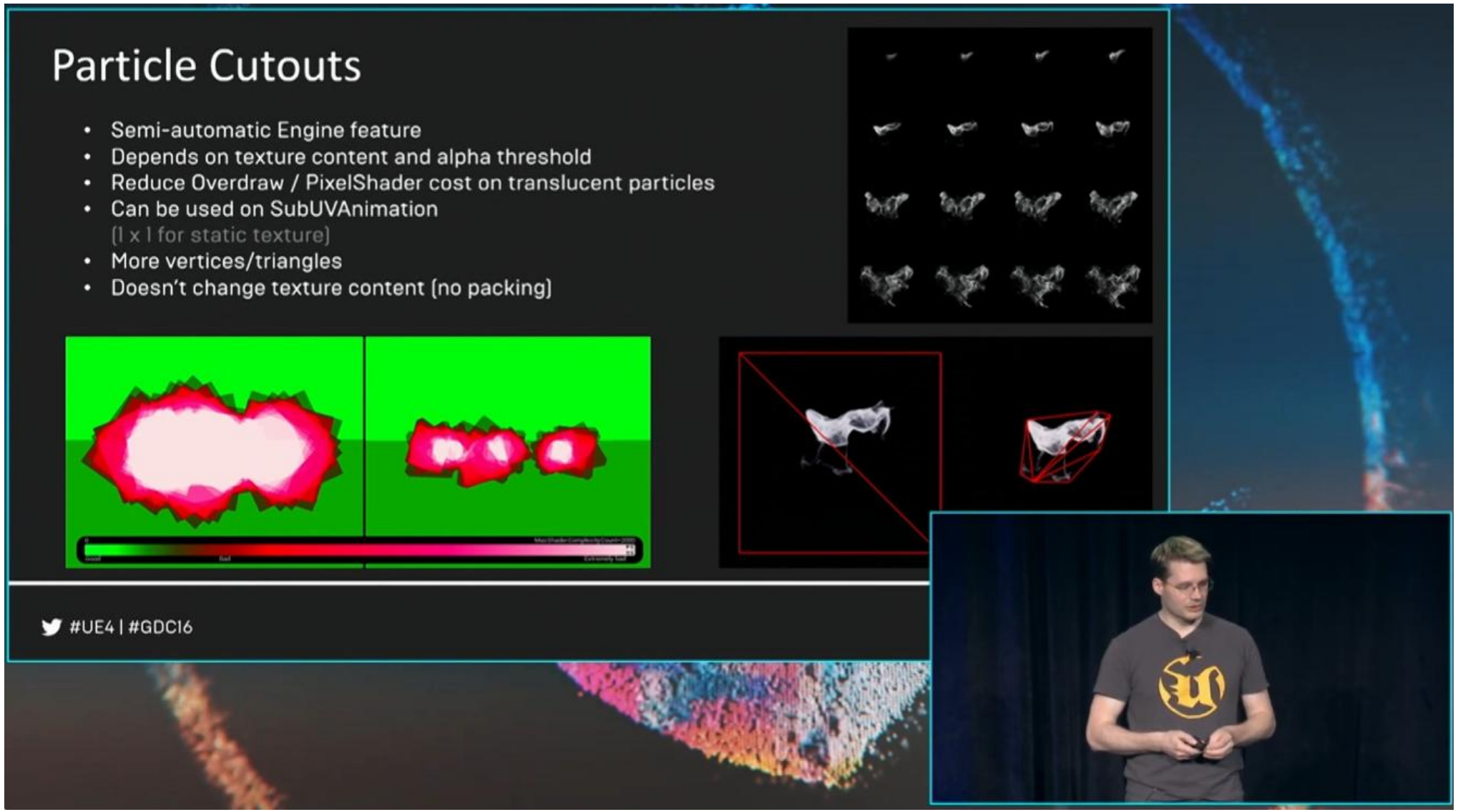


Final 8 vertex polygon (51.90%)

자료 출처 : Advanced in Real-Time Rendering in Games course – SIGGRAPH 2012 (https://advances.realtimerendering.com/s2012/Avalanche/Persson_GraphicsGemsForGames.pptx)

Cutout Texture

Building Paragon in UE4



자료 출처 : Building Paragon in UE4 | Epic Games live from GDC 2016 Event Coverage (https://youtu.be/BXcw2IrIinc?si=ek8EVsYExnDjZio3)

Flat billboard
Clouds Rendering

- "Real-Time Atmospheric Effects in Games Revisited"
  [Carsten Wenzel (CryTek) / Game Developers Conference, 2007]







자료 출처 : "Real-Time Atmospheric Effects in Games Revisited" | Presentation at Game Developers Conference 2007 (https://www.slideserve.com/kyna/real-time-atmospheric-effects-in-games)

World-Oriented based billboard

Soft Particle

- “DirectX-SDK-Samples10 – Direct3D10/Soft Particle”



자료 출처 : https://github.com/microsoft/DirectX-SDK-Samples

Soft Particle
Spherical Billboard

- “Spherical Billboards and Their Application to Rendering Explosions” [Tamas Umenhoffer et al / Graphics Interface, 2006]

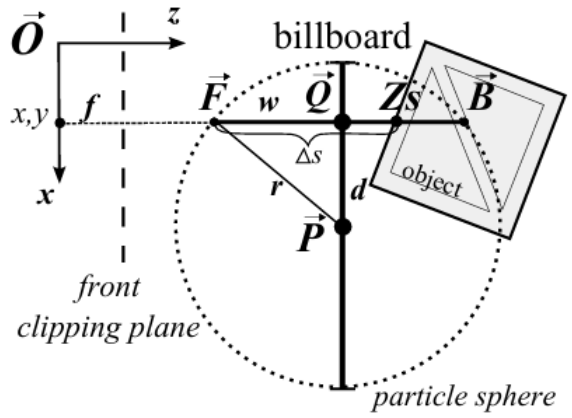




Planar



Spherical



자료 출처 : https://www.researchgate.net/publication/200019094_Spherical_billboards_and_their_application_to_rendering_explosions

Billboard

축 방향 빌보드(Axial billboard)

- 텍스처링된 오브젝트는 일반적으로 관측자를 똑바로 향하지 않음
- 대신 고정된 전역 공간 축을 중심으로 회전하고
  이 범위 내에서 가능한 한 관측자를 향하게 정렬할 수 있음
- 멀리 있는 나무를 표시하는 데 사용





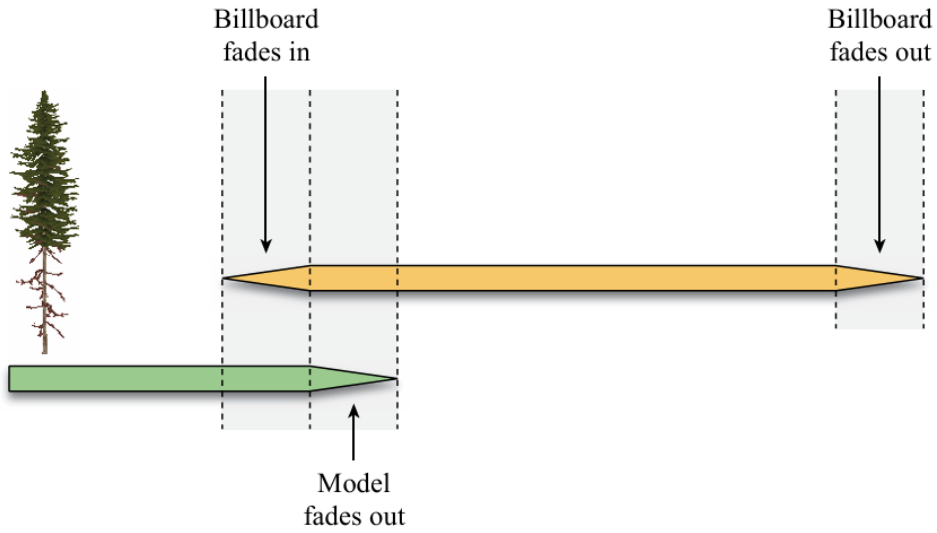
자료 출처 : Google Nano Banana 2

Axial Billboard

Vegetation Management

- "Vegetation Management in Leadwerks Game Engine 4" [Josh Klint / Game Engine Gems 3, 2016]

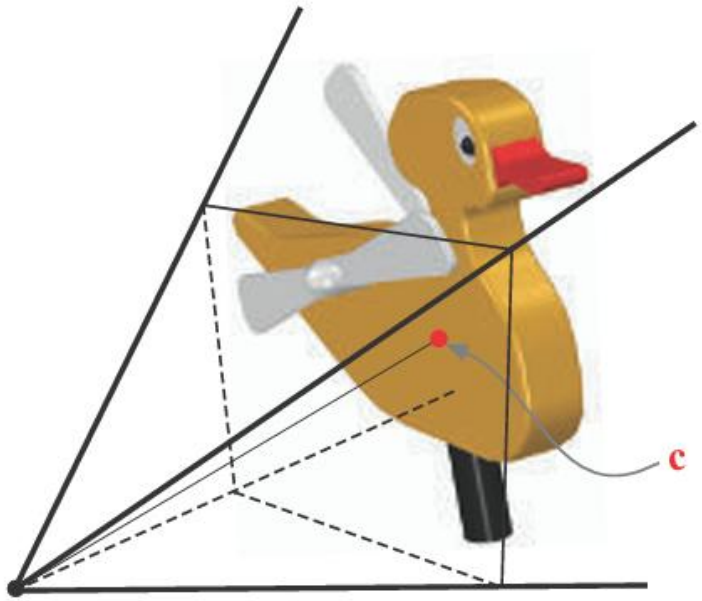


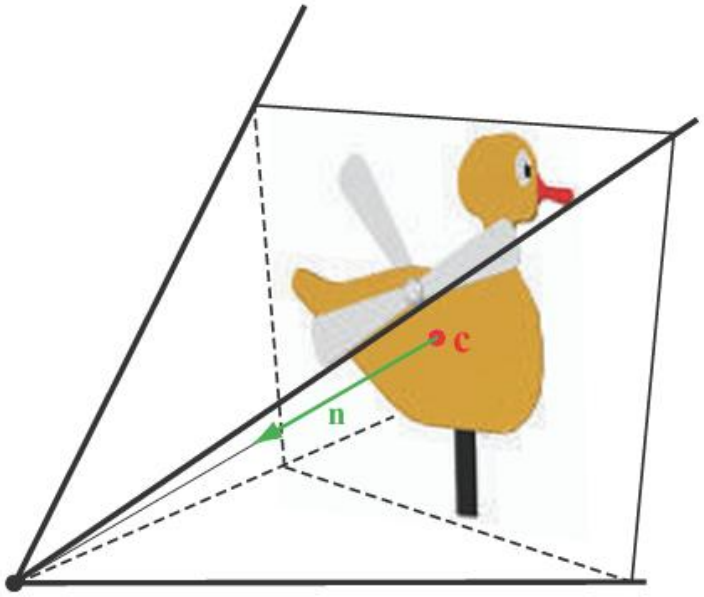


자료 출처 : https://youtu.be/FS_OiZ5kZ9Q?si=yTzQ08zrwBZQ9IE7 (Vegetation System LOD Preview), https://www.gameenginegems.net/geg3.php

Impostor

- Impostor 임포스터
  - n. (다른 사람 행세를 하는) 사기꾼
- “Imposters: Adding Clutter”
  [Tom Forsyth / Game Programming Gems 2, 2001]







자료 출처 : https://www.gamedeveloper.com/programming/imposters

Impostor (1/4)
Cloud Rendering (1/2)

- “Real-Time Cloud Rendering”
  [Mark J. Harris and Anselmo Lastra / Computer Graphics Forum, 2001]





자료 출처 : https://markmark.net/clouds/RTCRDownload.html

Impostor (1/4)
Cloud Rendering (2/2)

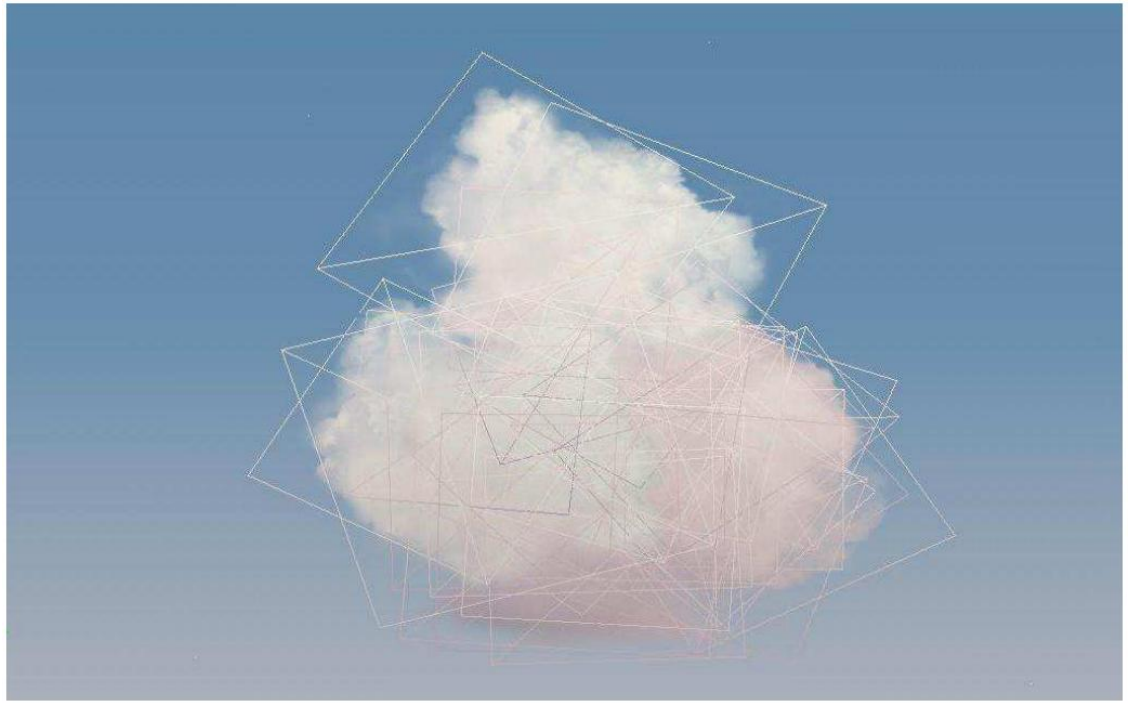
- "Realistic and Fast Cloud Rendering"
  [Niniane Wang / Journal of Graphics Tools, 2003]







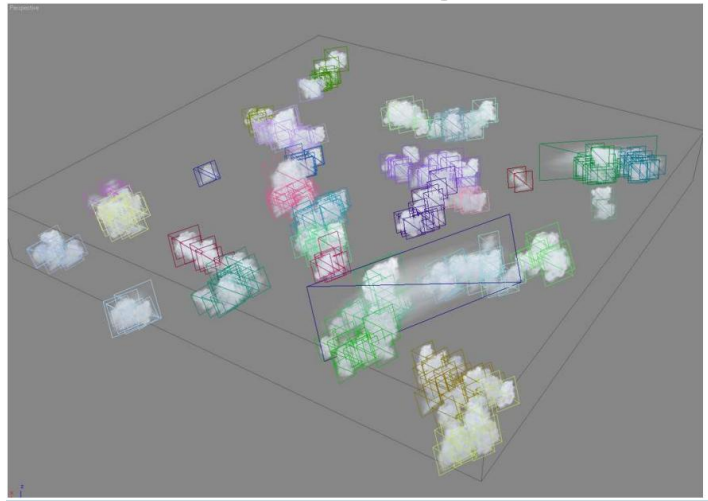


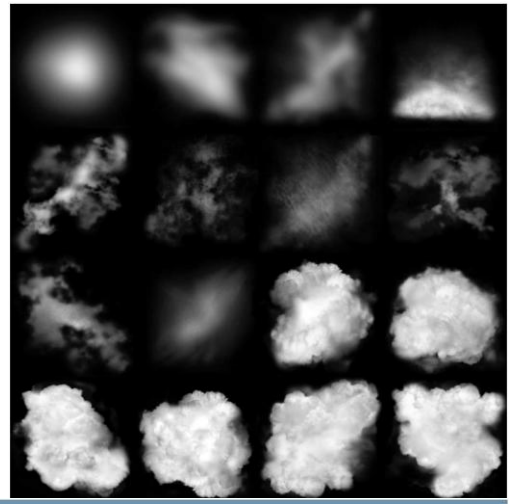


자료 출처 : https://niniane.org/clouds/

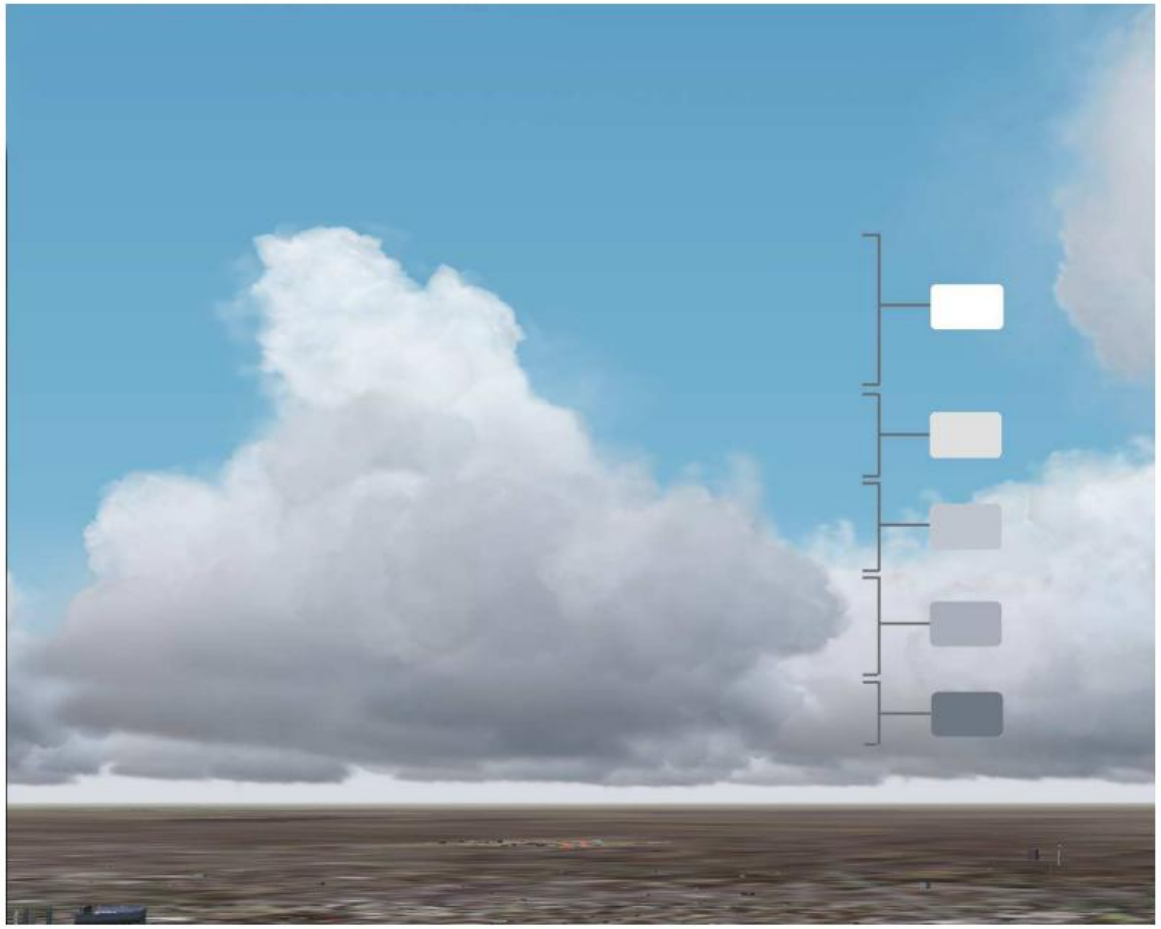
Impostor (1/4)
Cloud Rendering (2/2)

- "Realistic and Fast Cloud Rendering"
  [Niniane Wang / Journal of Graphics Tools, 2003]









자료 출처 : https://niniane.org/clouds/

Impostor (2/4)

- "Graphics Techniques in Crackdown"
  [Hugh Malan / ShaderX7 Chapter 2, 2009]



Clutter Off

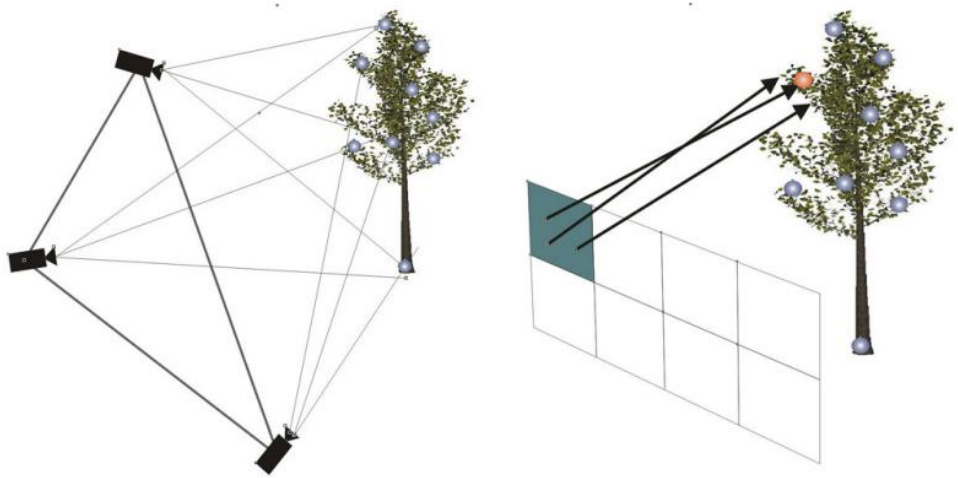


Clutter On

Impostor (3/4)

- "Point-Based Impostors for Real-Time Visualization"
  [Michael Wimmer *et al* / Rendering Techniques, 2001]





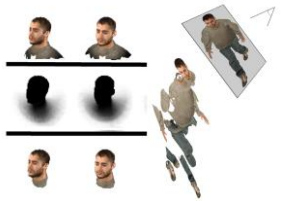


자료 출처 : https://inria.hal.science/inria-00510050/file/wimmer_egrw2001.pdf

Beyond Polygons

Impostor (4/4)





- "Polypostors: 2D Polygonal Impostors for 3D Crowds"
  [Ladislav Kavan *et al* / Interactive 3D Graphics and Games, 2008]
- "Efficient Rendering of Animated Characters through Optimzed per-joint Impostors"
  [A. Beacco *et al* / Journal of Computer Animation and Virtual Worlds, 2012]



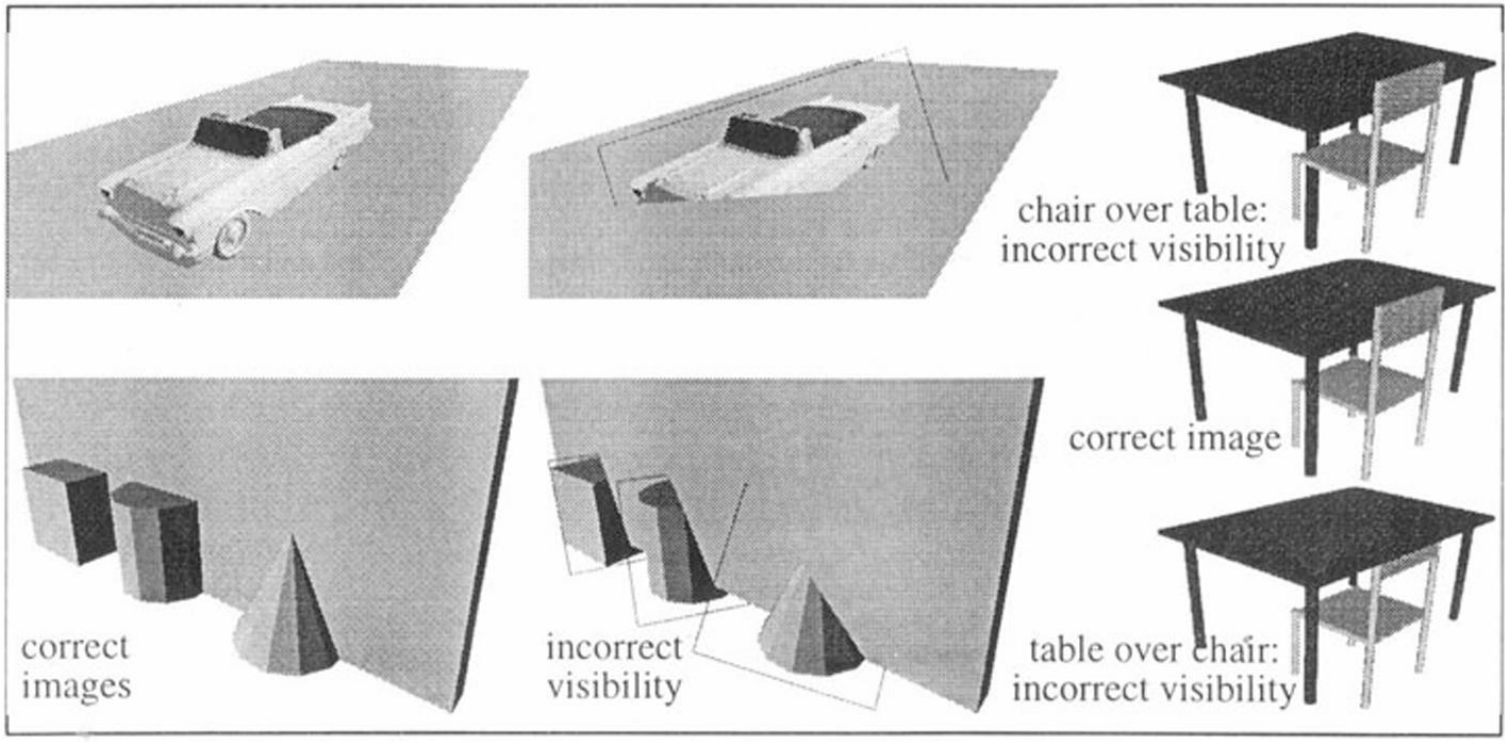


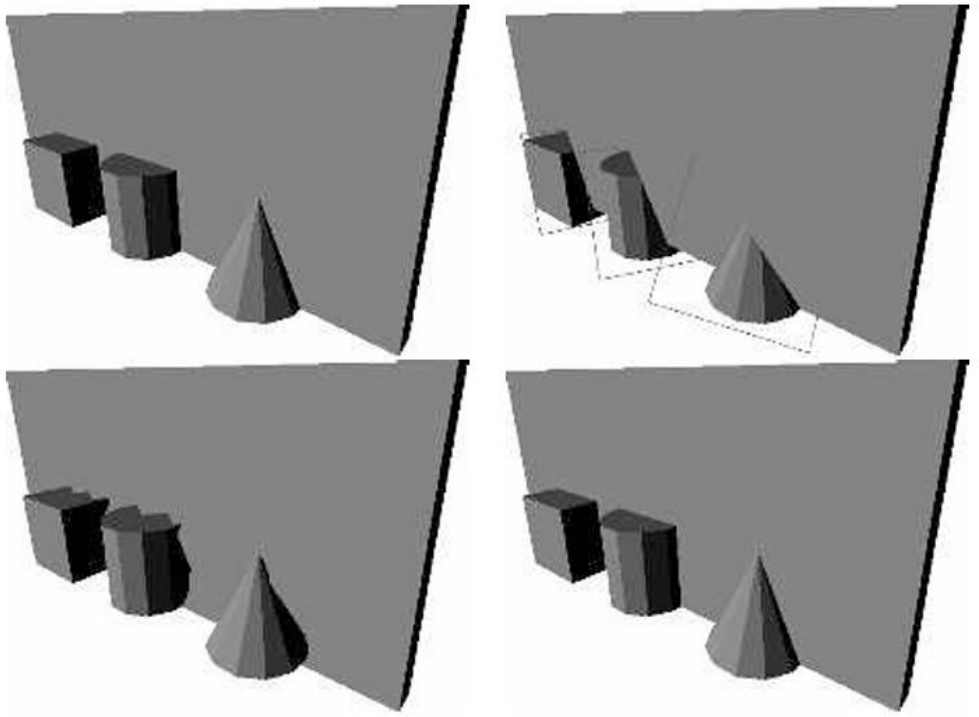
자료 출처 : https://youtu.be/DhtJf87yEFU?si=KF1RLvdZnF0fIaFV, // https://youtu.be/DyUdwEiULok?si=DuFBlJnhCj_8gsrN

Impostor Problem

Displacement Techniques (변위 기법)

- 렌더링된 이미지가 계속 관측자를 향해야 함
- 관측자를 향한 임포스터 자체에 두께가 없기 때문에 실제 지오메트리와 결합할 때 문제 발생
- “Nailboards: A Rendering Primitive for Image Caching in Dynamic Scenes” [Gernot Schaufler / Eurographics Workshop on Rendering, 1997]



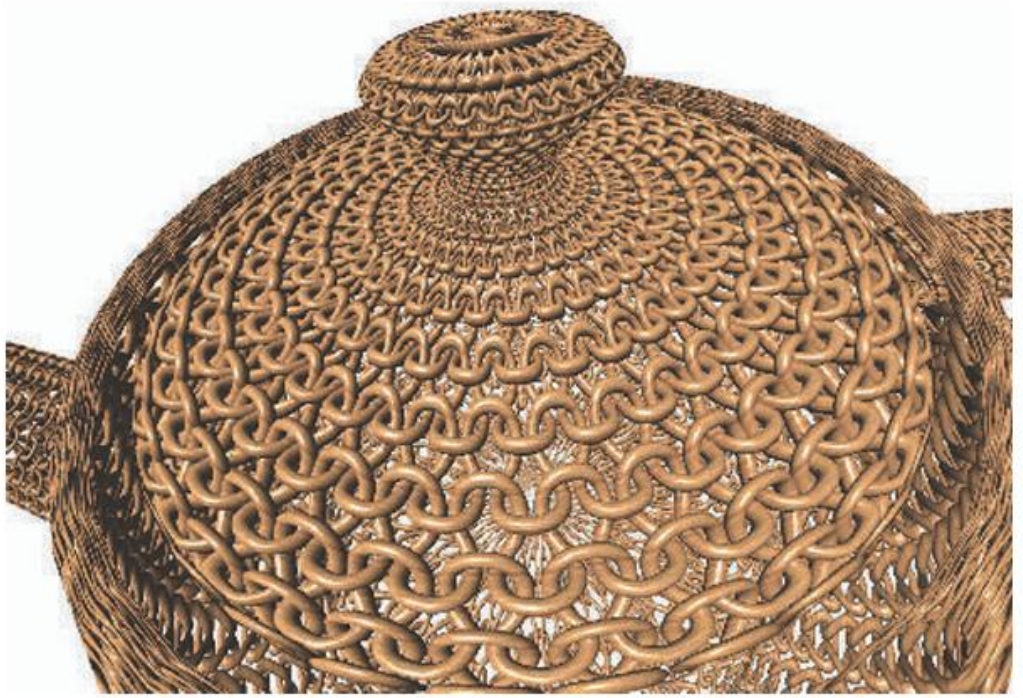


자료 출처 : https://link.springer.com/chapter/10.1007/978-3-7091-6858-5_14

릴리프 텍스처 매핑(Relief Texture Mapping)

- "Relief Texture Mapping"
  [Manuel M. Oliveira *et al* / SIGGRAPH, 2000]
- "Relief Mapping of Non-Height-Field Surface Details"
  [F. Policarpo & Manuel M. Oliveira / SIGGRAPH, 2006]



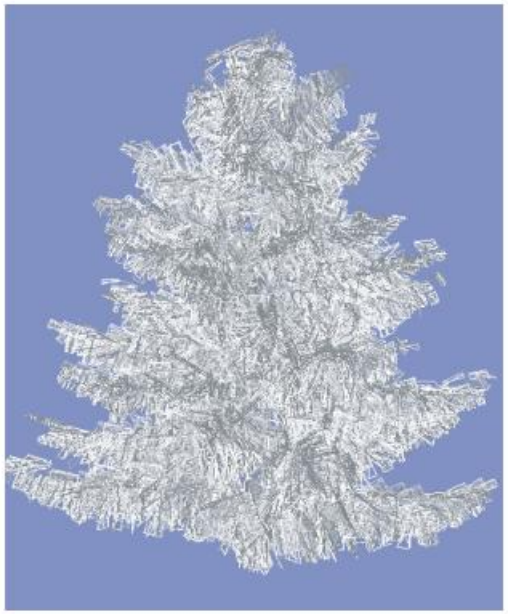


자료 출처 : https://vimeo.com/32003362

빌보드 구름(Billboard Cloud)

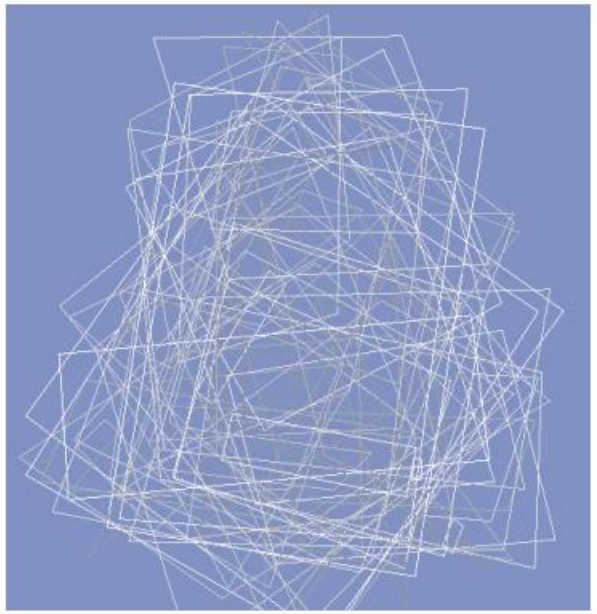
- "Stochastic Billboard Clouds for Interactive Foliage Rendering" [J. Dylan Lacewell *et al* / Journal of Graphics Tools , 2006]
  - 확률적 최적화 알고리즘을 이용해 가장 적은 수의 빌보드로 최대 입체감 표현





20,610 triangles 60FPS

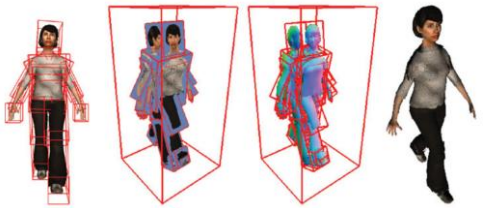




78 billboards 300FPS

자료 출처 : https://www.tandfonline.com/doi/abs/10.1080/2151237X.2006.10129213

빌보드 구름(Billboard Cloud)



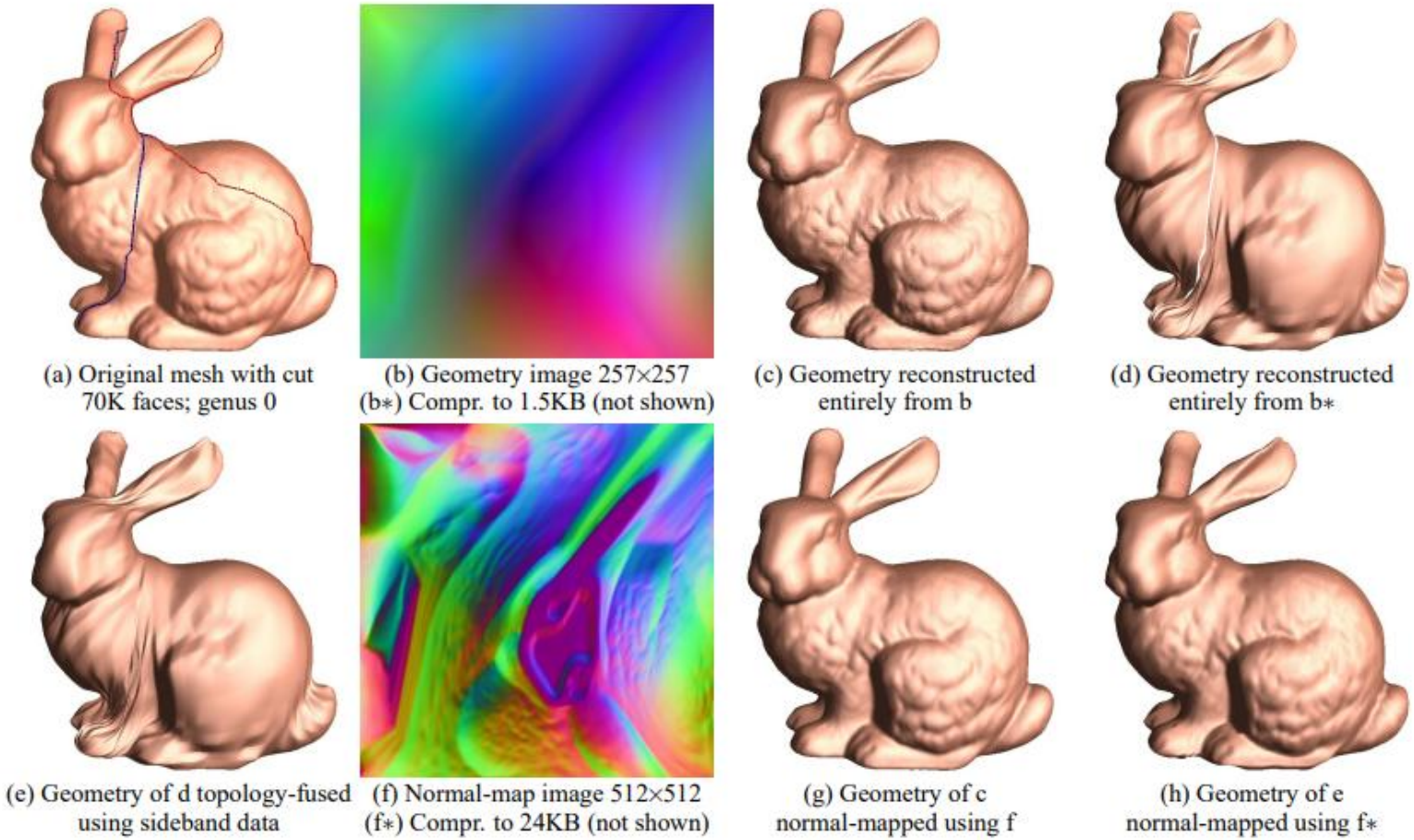
- "A Flexible Approach for Output-Sensitive Rendering of Animated Characters" [A. Beacco *et al* / Computer Graphics Forum, 2011]



자료 출처 : https://youtu.be/TjmpGX0D_vo?si=M7ABWZLa_0qpSrLV (A Flexible Approach for Output-Sensitive Rendering of Animated Characters – User Study)

Displacement Techniques
지오메트리 이미지

- "Geometry Images"
  [Xianfeng Gu *et al* / SIGGRAPH, 2002]

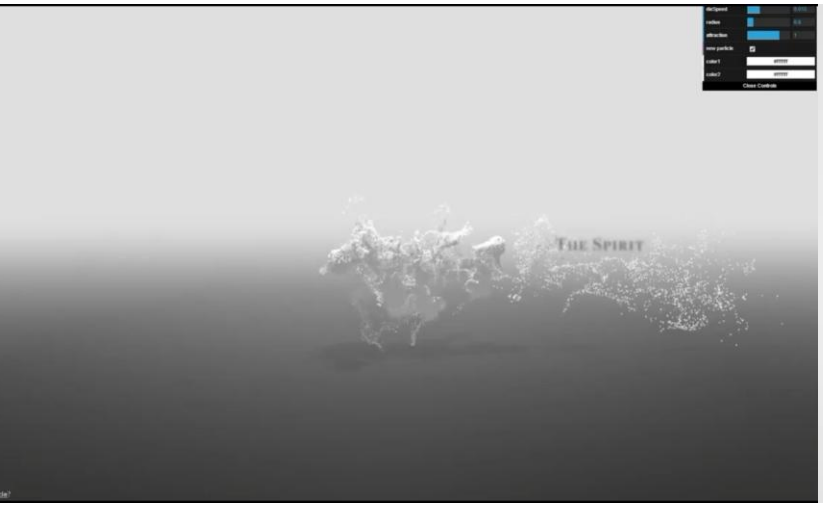




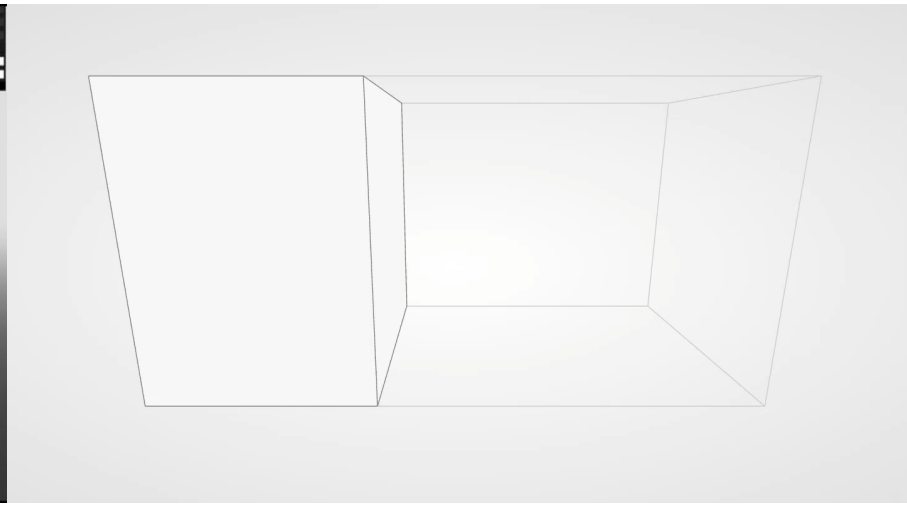
자료 출처 : glm buddha clip loop4 (https://youtu.be/XcvBqfS4Ea0?si=jxkL4OivjvHJxkQf)

Particle Systems

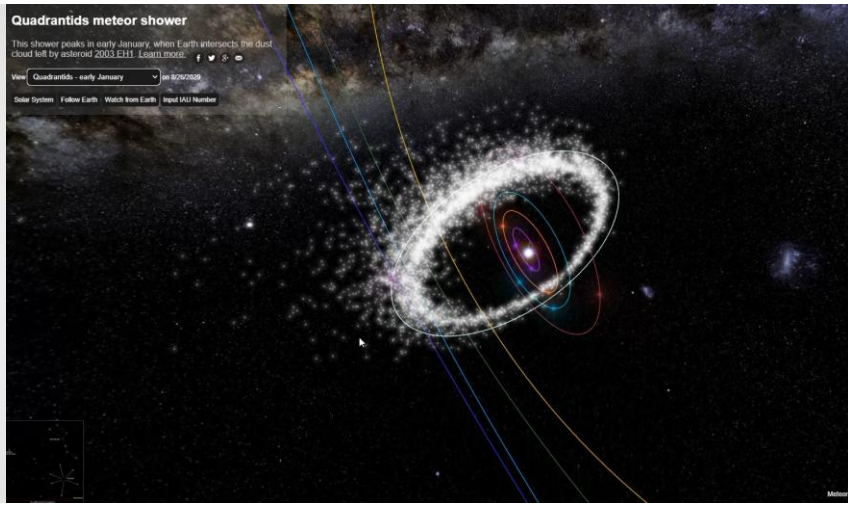
- 수 많은 작은 입자나 이미지를 대량으로 생성하고 애니메이션을 적용한 시스템
  - 각 입자는 단일 픽셀 또는 빌보드로도 표시
  - 입자가 둥글면 업 벡터는 디스플레이와는 무관
  - 각 입자에 대한 빌보드는 지오메트리 쉐이더 호출로 생성 가능
    - 실제로는 버텍스 쉐이더를 사용해서 스프라이트를 생성하는 것이 더 빠를 수 있음



https://edankwan.com/experiments/the-spirit/



https://david.li/fluid/



https://www.meteorshowers.org/

Optimizing the Rendering of a Particle System

- Ericson Christer 블로그 포스트 내용

- 파티클 데이터 및 에셋 최적화
  - 불투명 파티클 사용
  - 더 풍부한 파티클 스프라이트 사용
  - 투명 영역 줄이기

- 버퍼 및 해상도 트릭
  - AA-Resolve 이후에 파티클 그리기
  - **저해상도 버퍼 활용**
  - **MSAA 트릭으로 픽셀 쉐이더 연산 줄이기**
  - GPU에서 직접 파티클 생성

- GPU 및 파이프라인 최적화
  - 총 파티클 수 제한
  - 인스턴싱 및 데이터 공유
  - 상태 변경 최소화
  - 정렬 필요성 줄이기

- 고급/난해한 기법
  - 앞에서 뒤로(Front-to-back) 그리기
  - 파티클 객체 그룹화

저해상도 버퍼에 그려 병합하거나 MSAA를 해결한 후 렌더링

- "Destiny: From mythic Science Fiction to Rendering in Real-Time" [Natalya Tatarchuk *et al* / SIGGRAPH, 2013]



자료 출처 : https://advances.realtimerendering.com/s2013/index.html - PowerPoint Slides

저해상도 버퍼에 그려 병합하거나 MSAA를 해결한 후 렌더링

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자료 출처 : https://advances.realtimerendering.com/s2013/index.html - PowerPoint Slides

저해상도 버퍼에 그려 병합하거나 MSAA를 해결한 후 렌더링

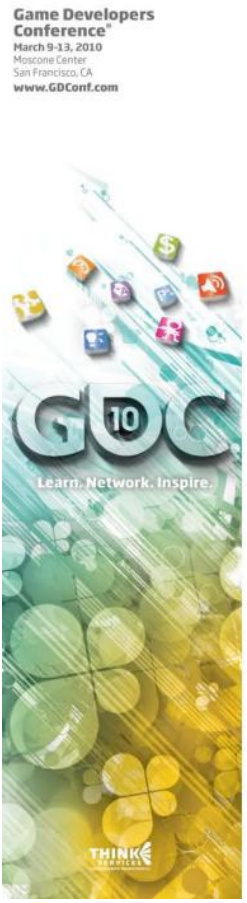
- "Destiny: From mythic Science Fiction to Rendering in Real-Time" [Natalya Tatarchuk *et al* / SIGGRAPH, 2013]

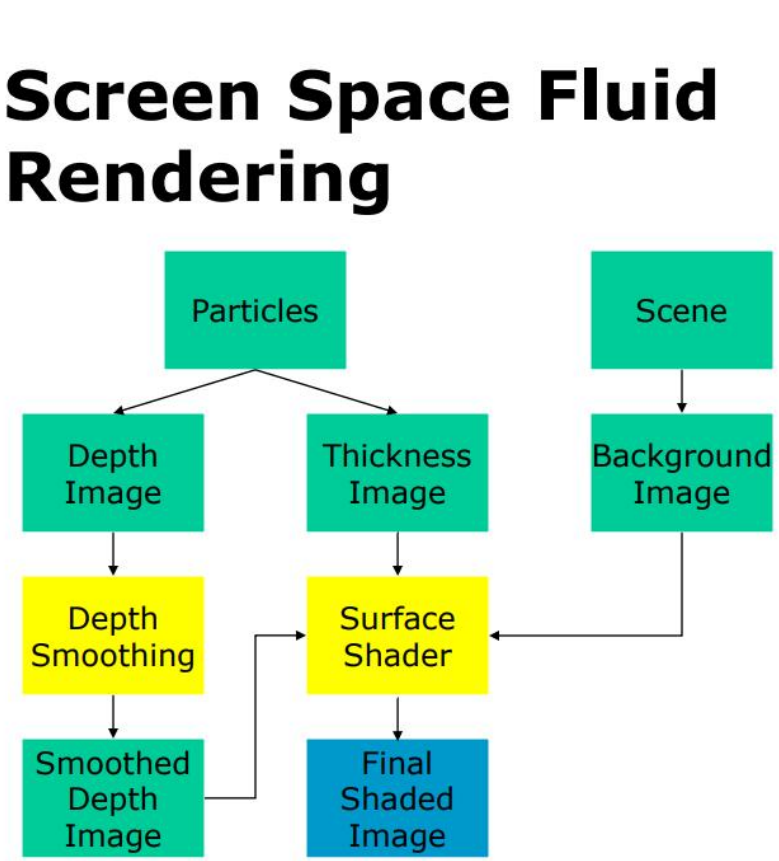


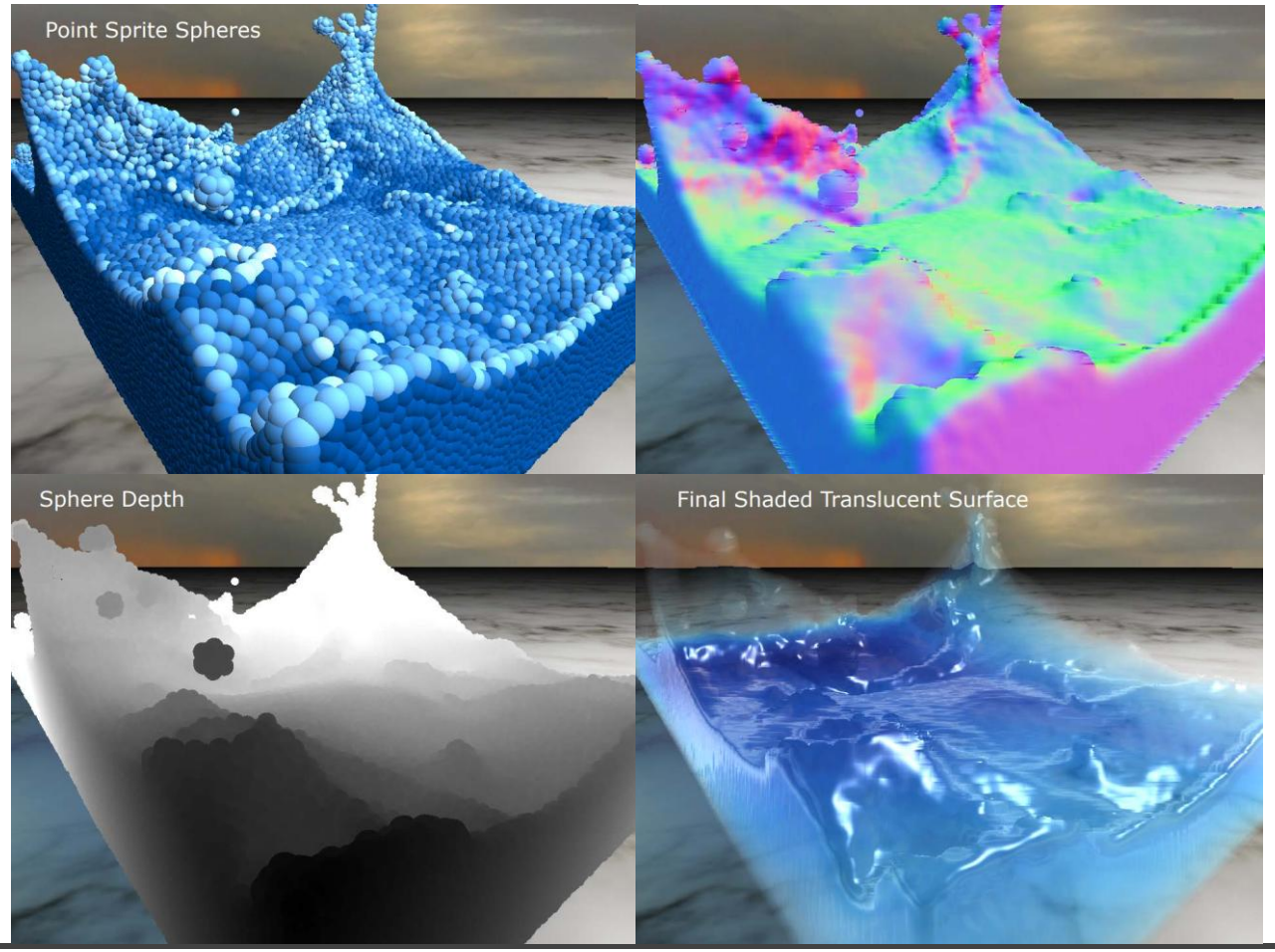
자료 출처 : https://advances.realtimerendering.com/s2013/index.html - PowerPoint Slides

Particle Systems
음영 입자

- "Screen Space Fluid Rendering for Games" [Simon Green / GDC, 2010]







자료 출처 : https://developer.download.nvidia.com/presentations/2010/gdc/Direct3D_Effects.pdf

Shading Particles

Volumetric Particle Lighting

- "Shiny PC Graphics in Battlefield 3"
  [Johan Andersson / GeForce LAN 6, 2011]

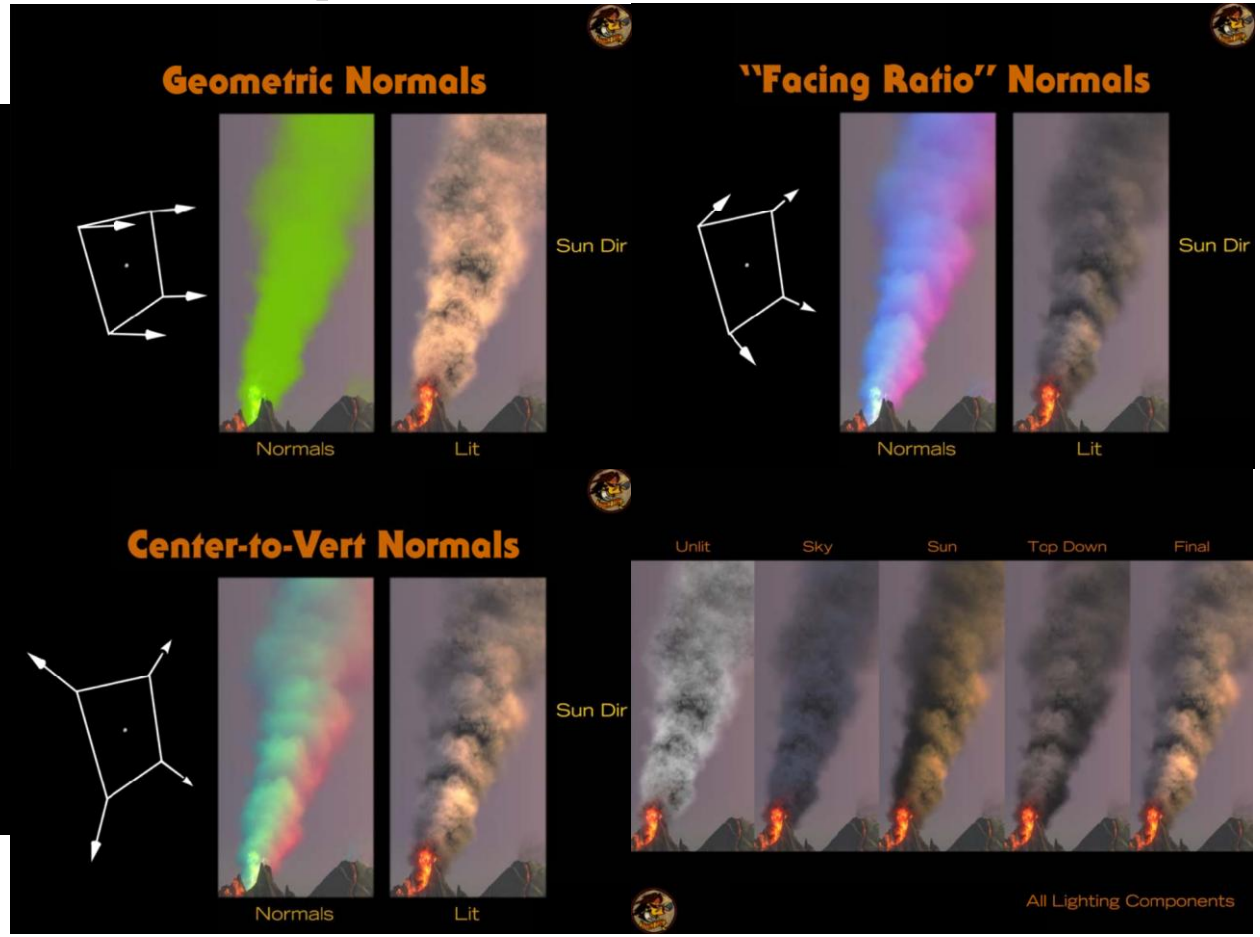


자료 출처 : https://youtu.be/EN549g0GNgc?si=4sCzbgP5R9RemCij (GeForce LAN 6 – Johan Andersson Battlefield 3 Presentation Part 3 / 5)

Optimizing Particle Lighting Using Vertex Normal Tricks

- "Rock Show VFC: Bringing Brutal Legend to Life"
  [Drew Skillman and Peter Demoreuille / GDC, 2010]



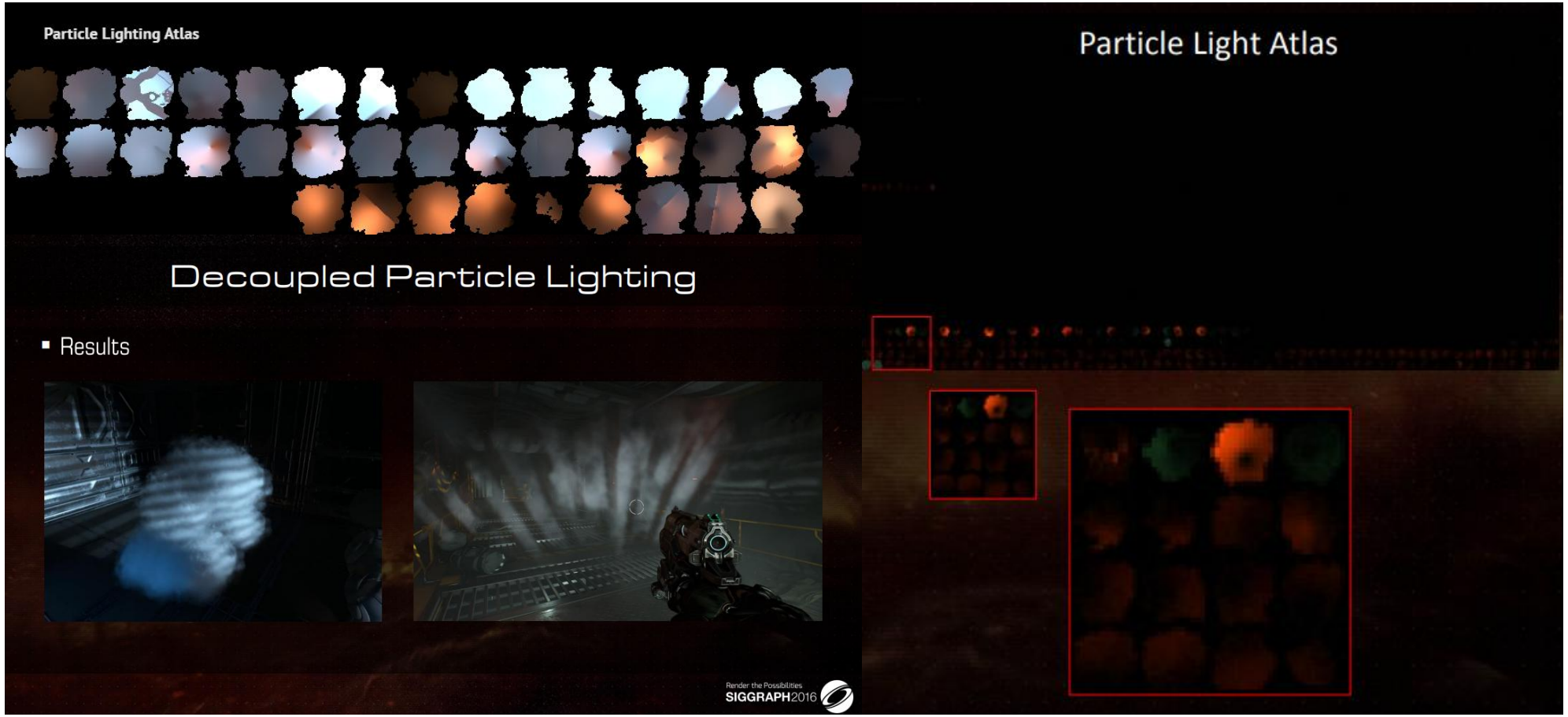


자료 출처 : https://youtu.be/XkvykLMr9eI?si=0S9d8vBnu0sG-55U (Brutal Legen Time Lapse Climate Video), https://projects.drewskillman.com/assets/downloads/GDC2010_VFX.pdf

Shading Particles

Transparents & Particle Lighting

- “Doom: The Devil is in the Details”
  [Tiago Sousa and Jean Geffroy / SIGGRAPH, 2016]



자료 출처 : https://advances.realtimerendering.com/s2016/Siggraph2016_idTech6.pdf

FOM(Fourier Opacity Mapping)

- "Fast Rendering of Opacity-Mapped Particles Using DirectX 11 Tessellation and Mixed Resolutions" [Jon Jansen and Louis Bavoil / NVIDIA White Paper, 2011]



Ground Truth



Vertex Lighting

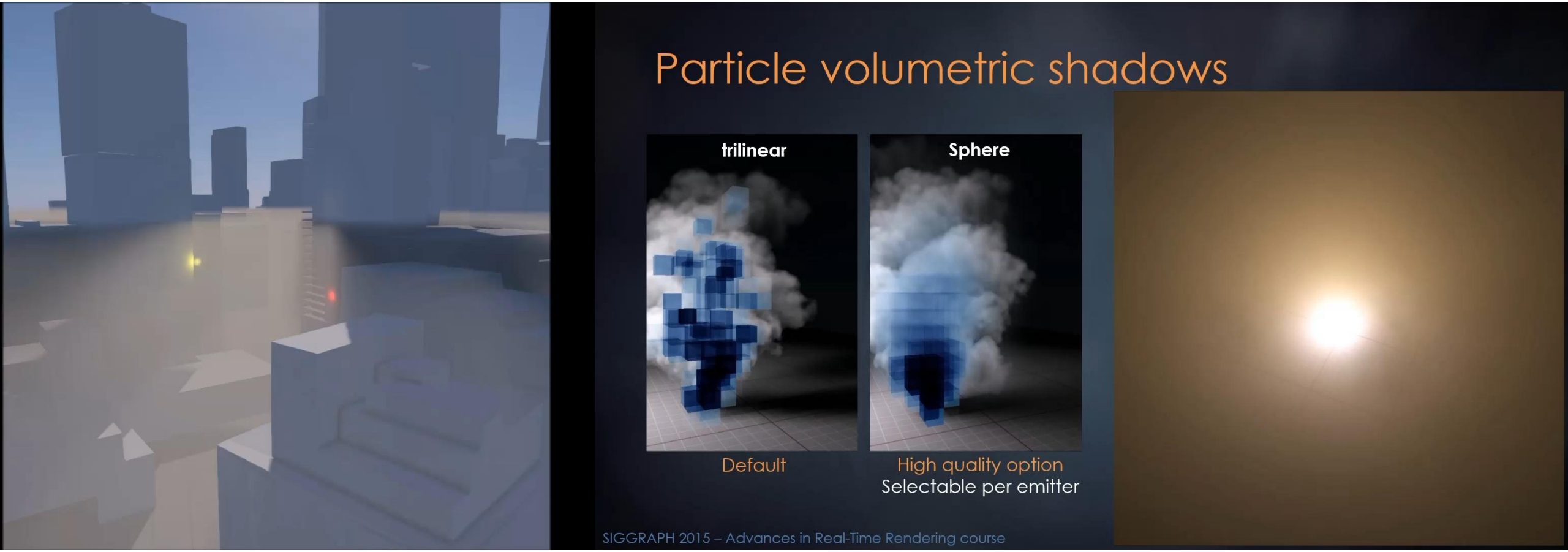


Tessellated Lighting

자료 출처 : https://developer.download.nvidia.com/assets/gamedev/files/sdk/11/OpacityMappingSDKWhitePaper.pdf

Shading Particles
Volumetric Fog

- “Physically-based and Unified Volumetric Rendering in Frostbite” [Sebastien Hillaire / SIGGRAPH, 2015]



자료 출처 : https://www.slideshare.net/slideshow/physically-based-and-unified-volumetric-rendering-in-frostbite/51840934

Particle Simulation

- "The Destiny Particle Architecture"
  [Brandon Whitley / SIGGRAPH, 2017]

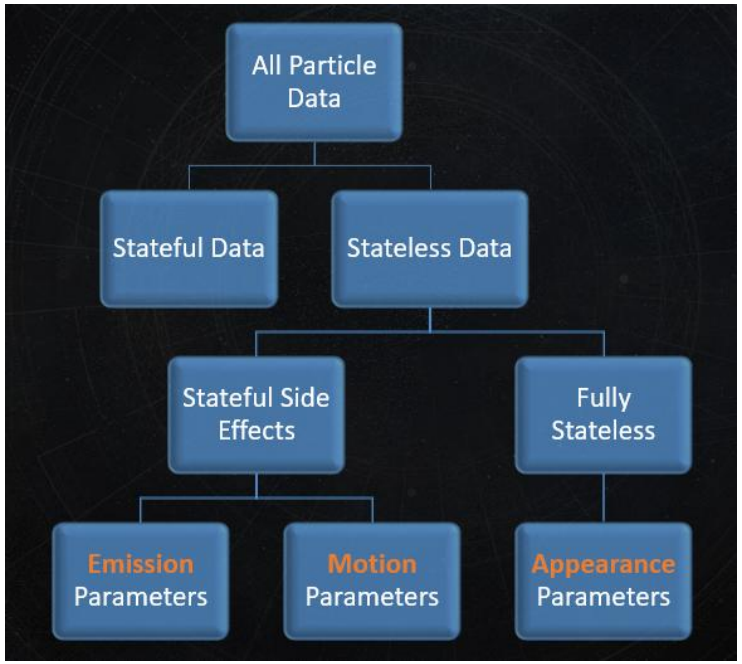


자료 출처 : https://advances.realtimerendering.com/s2017/index.html

Particle Simulation

The Destiny Particle Architecture (1/2)

- 파티클 수식의 실시간 해석을 위해 자체적인 TFX 바이트코드 가상 머신 설계
  - 데이터 단위 (128-bit SIMD)
  - 초경량 명령어 (Lean Opcodes)
  - 제어 흐름의 배제 (분기문이나 루프 같은 제어 흐름 명령어 없음)
- 하이브리드 파티클 코드 아키텍처 (CPU-GPU 통합 파이프라인)
  - CPU 데이터 구조
    - 바이트코드 실행
    - SIMD 기반 연산
  - GPU Compute Shader (HLSL)
    - 쉐이더 컴파일러 변환
    - 전용 스토리지 버퍼 바인딩
- 작업 그래프(Job Graph) 기반 멀티스레딩 프레임워크
  - 모든 연산을 잘게 쪼갠 태스크 중심의 작업 그래프 구조



자료 출처 : https://advances.realtimerendering.com/s2017/index.html

Particle Simulation

The Destiny Particle Architecture (2/2)

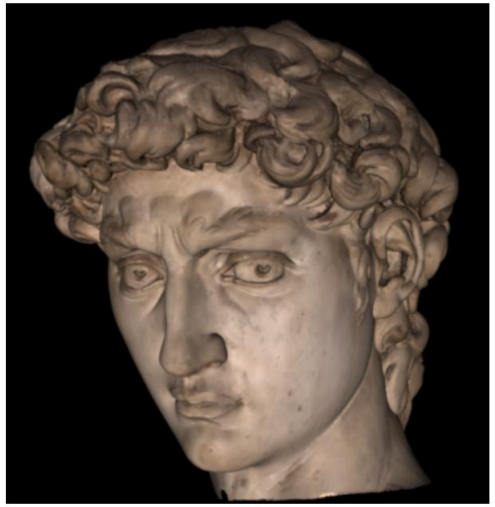


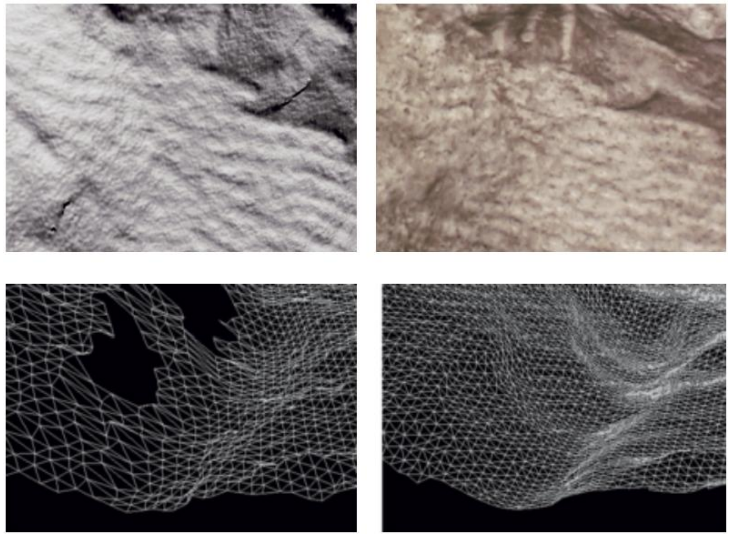
자료 출처 : https://advances.realtimerendering.com/s2017/index.html

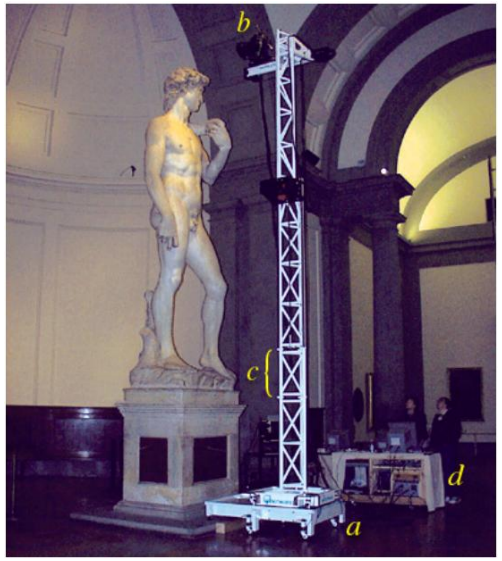
Point Rendering (1/2)

- "The Use of Points as a Display Primitive"
  [Marc Levoy and Turner Whitted / University of North Carolina, 1985]
  - 모든 것을 렌더링하는 데 사용할 새로운 기본 요소로 점(Point)을 사용할 것을 제안
  - 이유 : 어차피 최종 화면(픽셀)에 표시될 때 아주 작아질 폴리곤이라면,
    처음부터 수학적 면적을 가지지 않는 점(Point)으로 표현하는 것이 훨씬 효율적
- "The Digital Michelangelo Project: 3D Scanning of Large Statues"
  [Marc Levoy *et al* / SIGGRAPH, 2000]





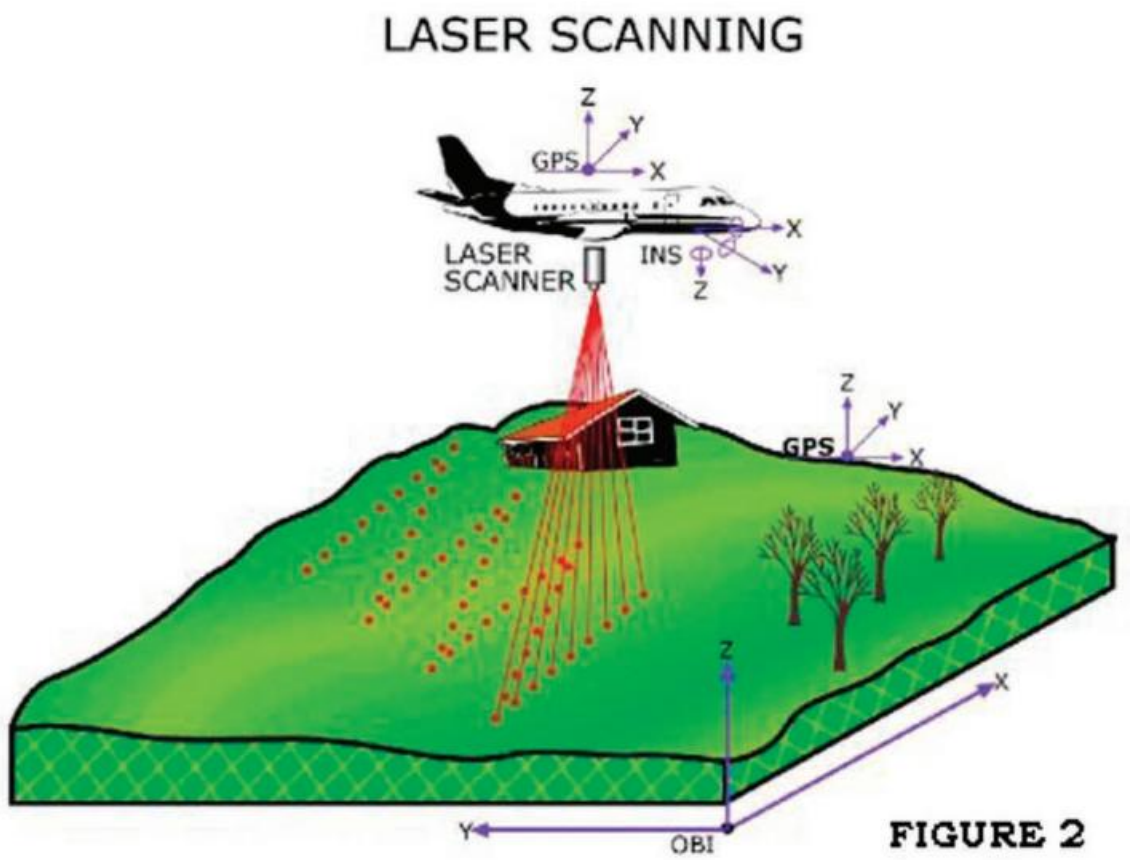


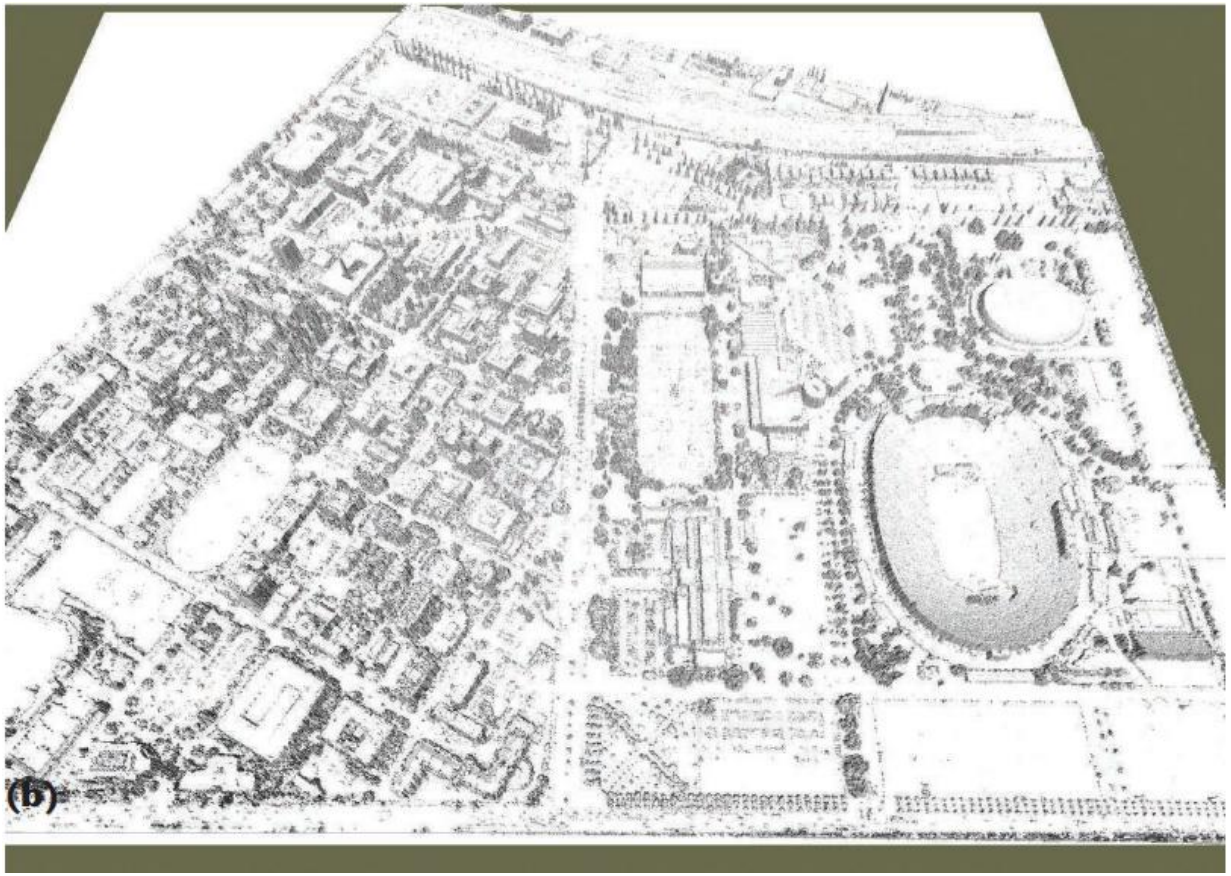


자료 출처 : https://dl.acm.org/doi/epdf/10.1145/344779.344849

Point Rendering (2/2)

- "Approaches to Large-Scale Urban Modeling"
  [Hu Jinhui *et al* / IEEE Computer Graphics and Applications, 2003]





자료 출처 : https://www.computer.org/csdl/magazine/cg/2003/06/mcg2003060062/13rRUwh80Nq

Point-based Renderer

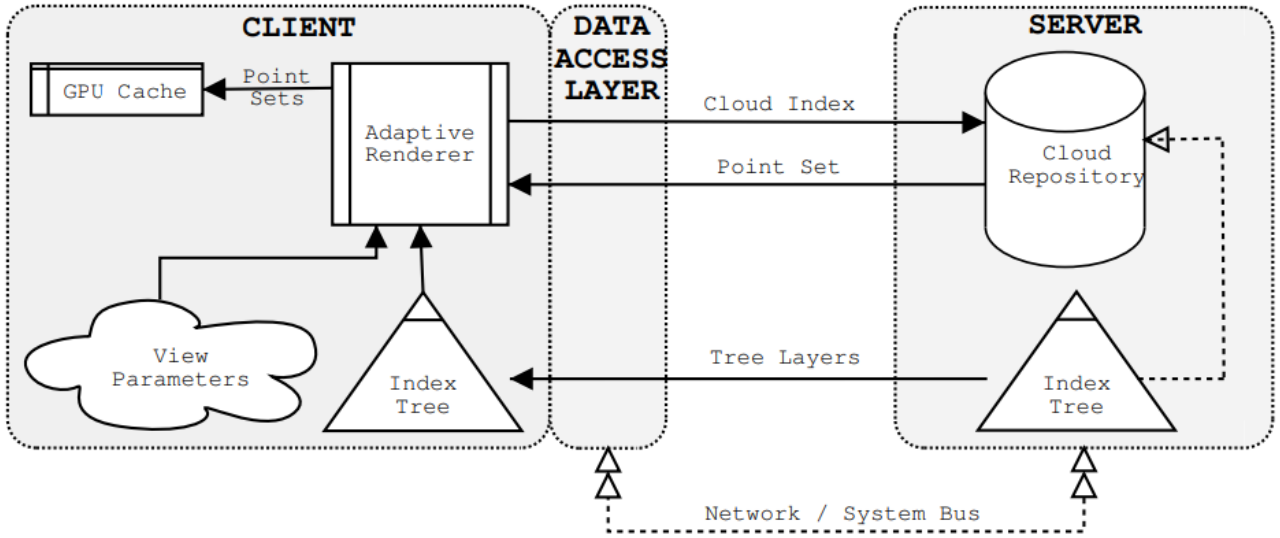
- "QSplat: A Multiresolution Point Rendering System for Large Meshes" [Szymon Rusinkiewicz and Marc Levoy / SIGGRAPH, 2000]



자료 출처 : https://dl.acm.org/doi/epdf/10.1145/344779.344940

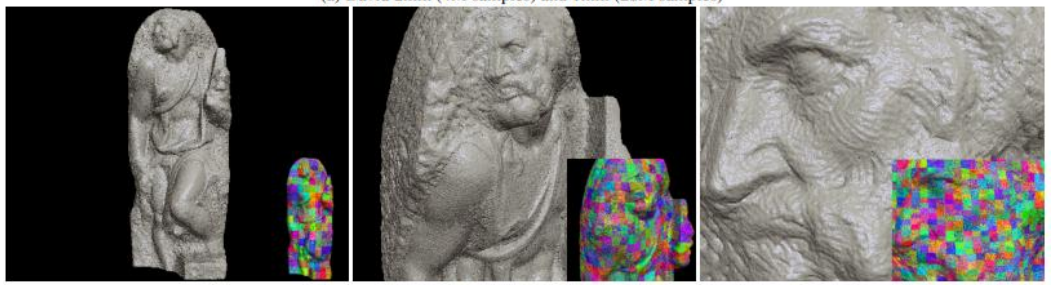
Layered Point Clouds

- "Layered Point Clouds"
  [Enrico Gobbetti and Fabio Marton / Symposium on Point-Based Graphics, 2004]

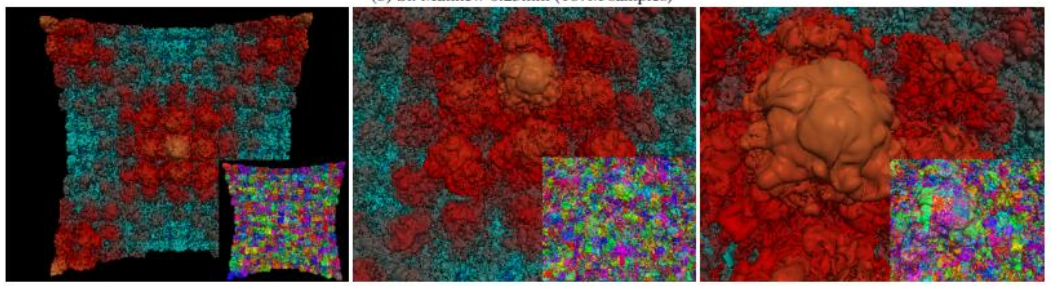




(a) David 2mm (4M samples) and 1mm (28M samples)



(b) St. Matthew 0.25mm (167M samples)



(c) Mixing interface isosurface (234M samples)

자료 출처 : https://dl.acm.org/doi/epdf/10.1145/344779.344940

Point Cloud Rendering

- "Potree: Rendering Large Point Clouds in Web Browsers"
  [Markus Schuetz / Diploma thesis in Visual Computing of Vienna University, 2016]



자료 출처 : http://potree.org/potree/examples/cesium_retz.html

Point Rendering
Splatting

- “Learning from failure: A Survey of Promising, Unconventional and Mostly Abandoned Renderers for ‘Dreams PS4’”
  [Alex Evans / SIGGRAPH, 2015]



자료 출처 : https://youtu.be/4j8Wp-sx5K0?si=ELS7cDrLO4x2jZaX (DREAMS –E3 2015 | PS4)

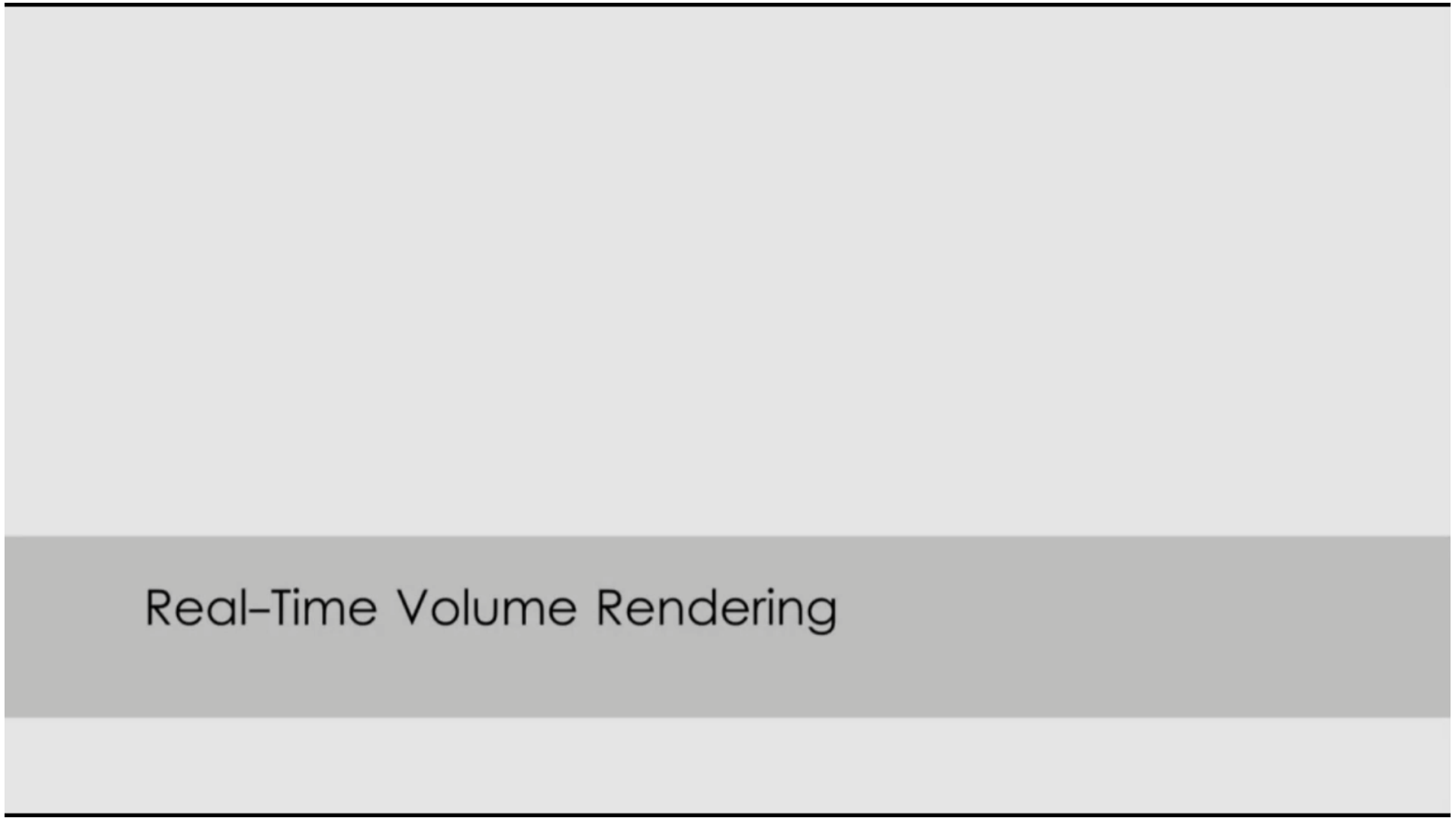
Beyond Polygons

Voxels

- 픽셀(Pixel) – 이미지 요소(Picture Element)
- 텍셀(Texel) – 텍스처 요소(Texture Element), Texture + Pixel
- 복셀(Voxel) – 볼륨 요소(Volume Element), Volume + Pixel
  - 각 복셀은 균일한 3차원 격자에서 공간의 볼륨을 나타냄
  - 입체적인 물체의 단면 표시 및 부분 재구성이 가능
    - 공간 전체에 대한 특성값을 기록할 수 있기 때문에
  - 격자의 인덱스가 위치를 결정하기 때문에 복셀당 위치 정보는 필요없음

GPU-based Voxel Database

- "GVDB: Raytracing Sparse Voxel Database Structures on the GPU" [Rama Karl Hoetzlein / High Performance Graphics, 2016]

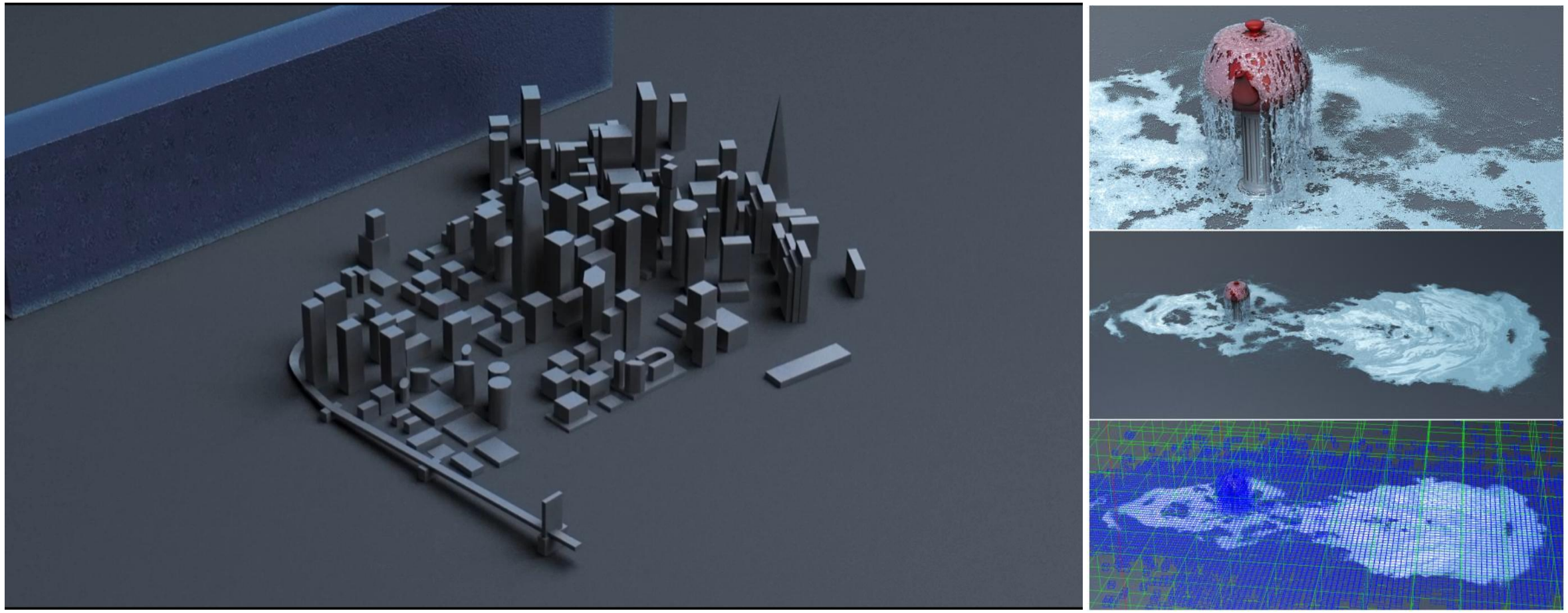




자료 출처 : https://youtu.be/7VPPKKpde3A?si=MOv0bGRipIwngin4 ([EG 2018] Fast Fluid Simulations with Sparse Volumes on the GPU)

Voxel-based Fluid Simulation

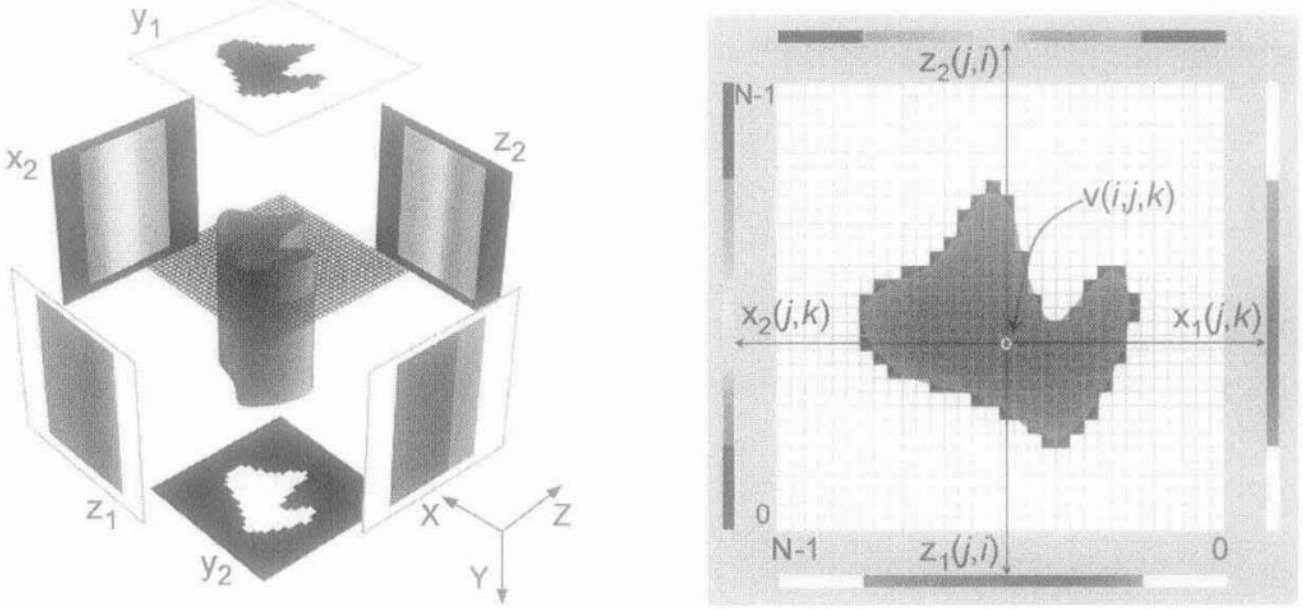
- "Fast Fluid Simulations with Sparse Volumes on the GPU" [Kui Wu *et al* / Eurographics, 2018]



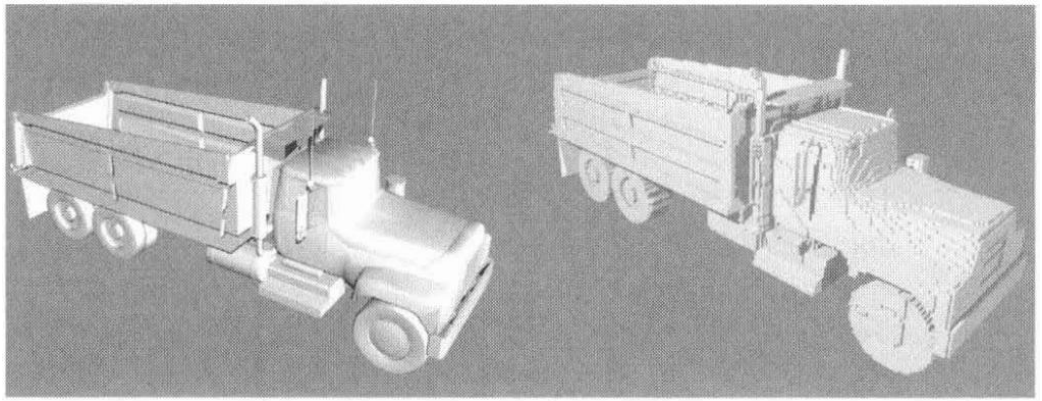
자료 출처 : https://youtu.be/7VPPKKpde3A?si=MOv0bGRipIwngin4 ([EG 2018] Fast Fluid Simulations with Sparse Volumes on the GPU)

Voxel Storage (1/2)

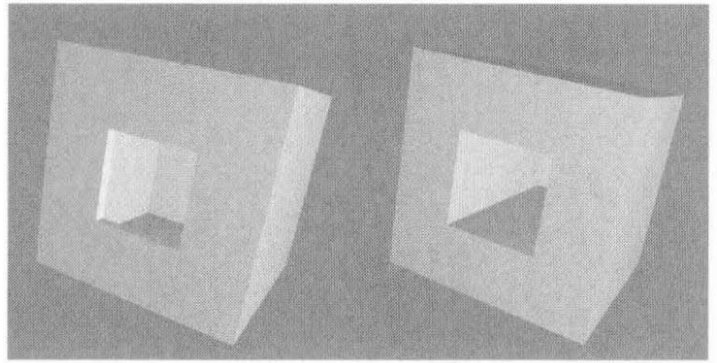
- "A Fast Depth-Buffer-Based Voxelization Algorithm"
  [Evaggelia-Aggeliki Karabassi *et al* / Journal of Graphics Tools, 1999]



Voxelization of an object using a 32x32x32 voxel space



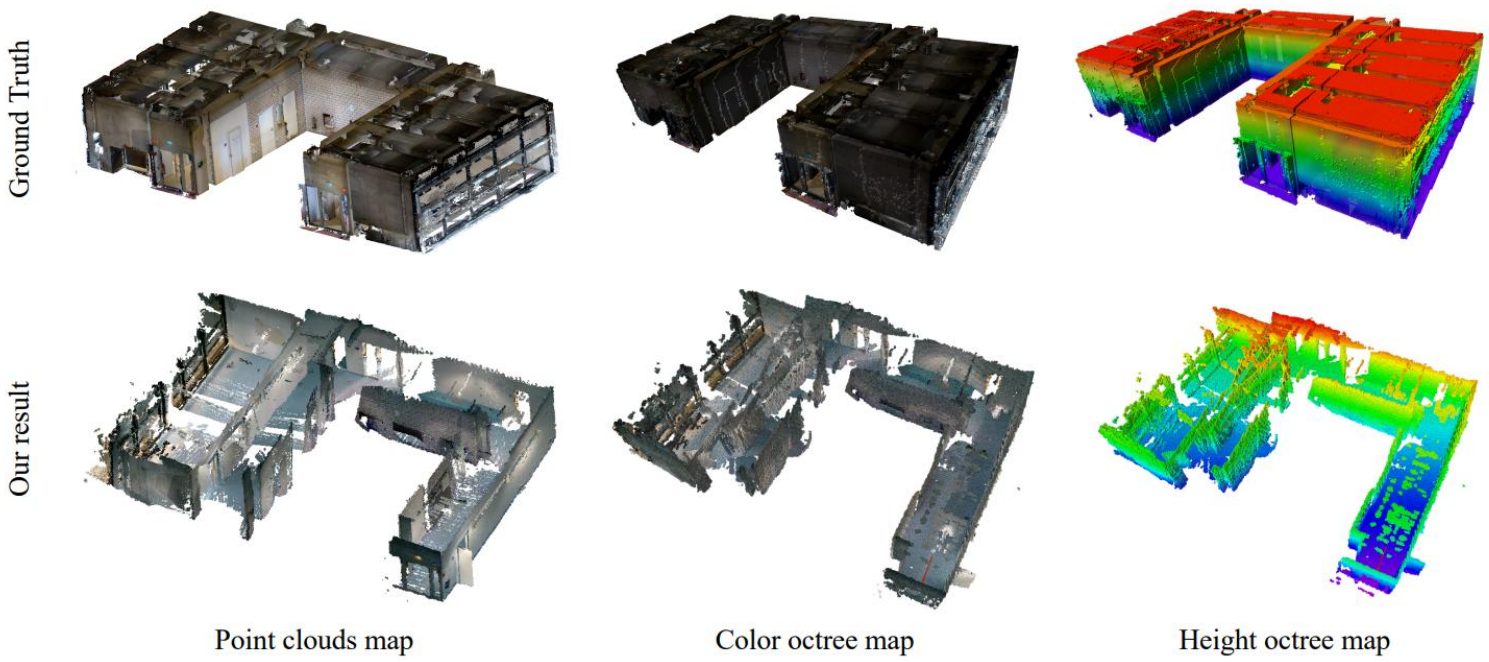
Truck voxelized

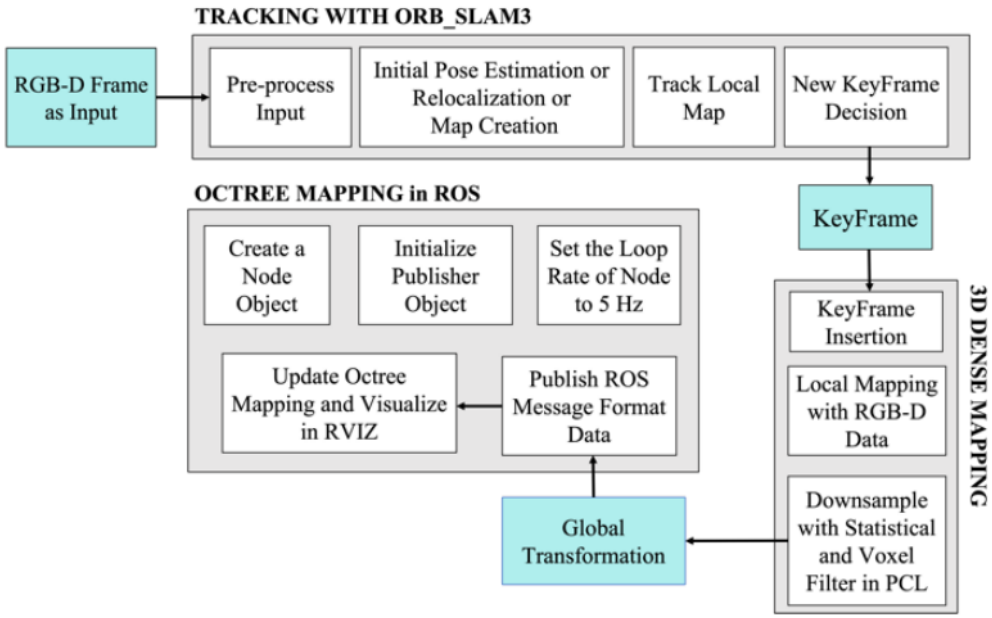


Error in voxelization

Voxel Storage (2/2)

- "Octree-Based Approach for Real-Time 3D Indoor Mapping Using RGB-D Video Data" [Jiwei Hou *et al* / ISPRS, 2023]



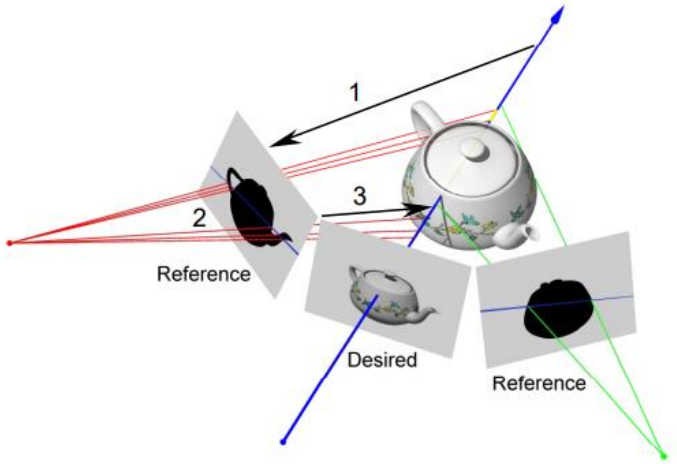


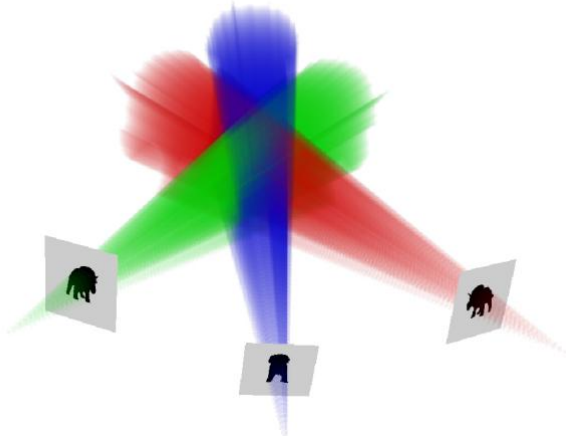
자료 출처 : https://isprs-archives.copernicus.org/articles/XLVIII-1-W1-2023/183/2023/isprs-archives-XLVIII-1-W1-2023-183-2023.pdf

Image-Based Visual Hulls

- "Image-Based Visual Hulls"
  [Wojciech Matusik *et al* / SIGGRAPH, 2000]



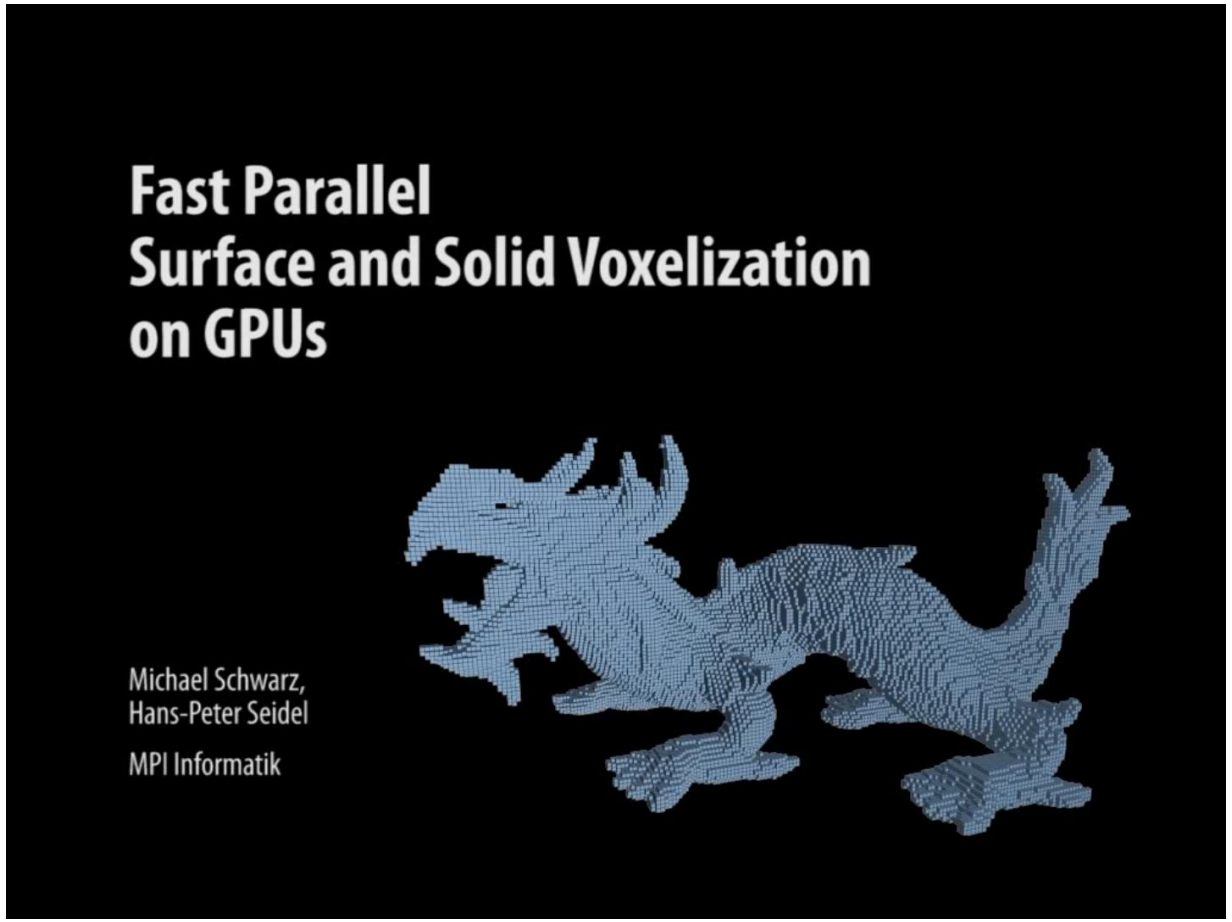


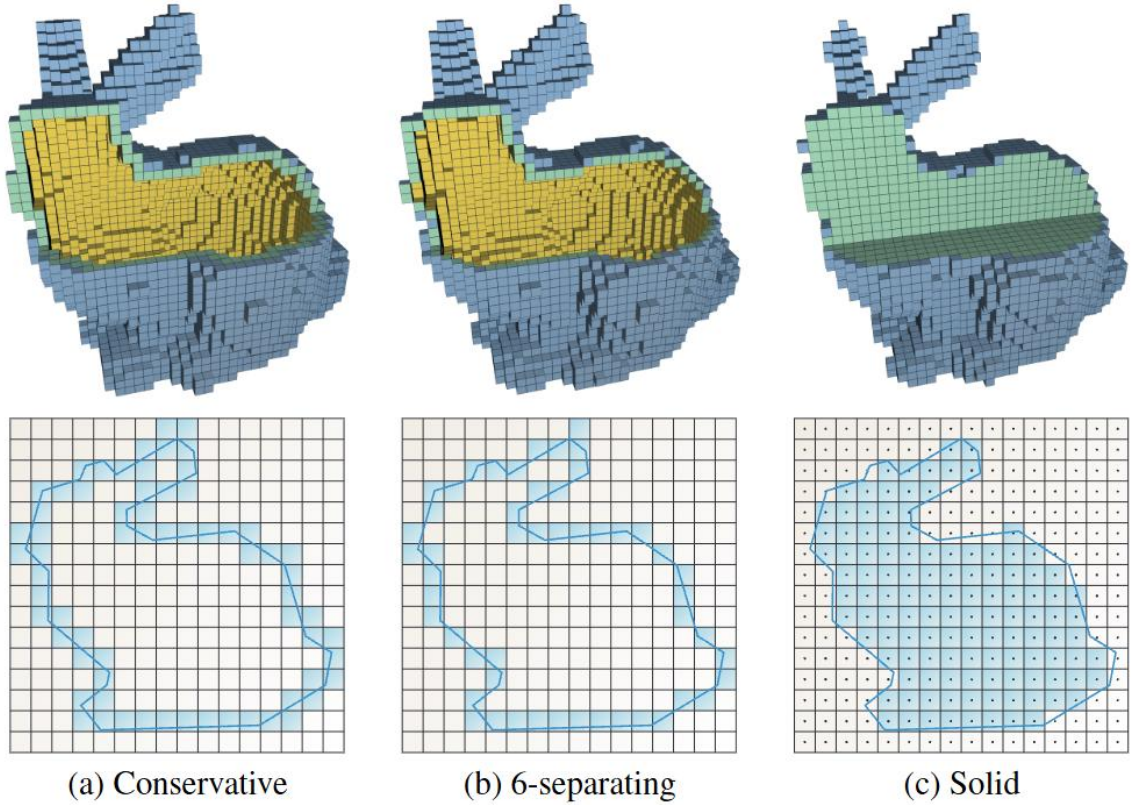


자료 출처 : https://www.eecs.harvard.edu/~sjg/papers/ibvh.pdf

Voxel Storage
Voxelization

- “Fast Parallel Surface and Solid Voxelization on GPUs” [Michael Schwarz and Hans-Peter Seidel / SIGGRAPH Asia, 2010]

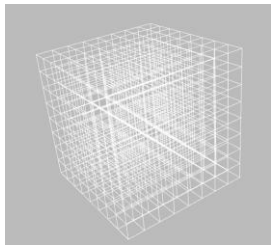




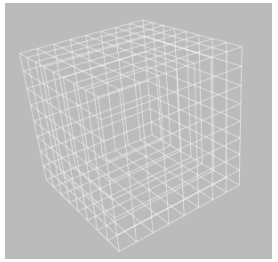
자료 출처 : https://michael-schwarz.com/research/publ/2010/vox/

Voxel Rendering

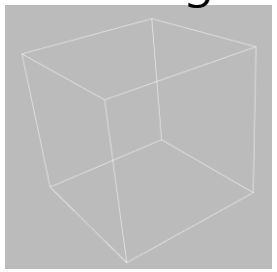
- "Meshing in a Minecraft Game"
  [Lysenko Mikola / 0 FPS blog, 2012]



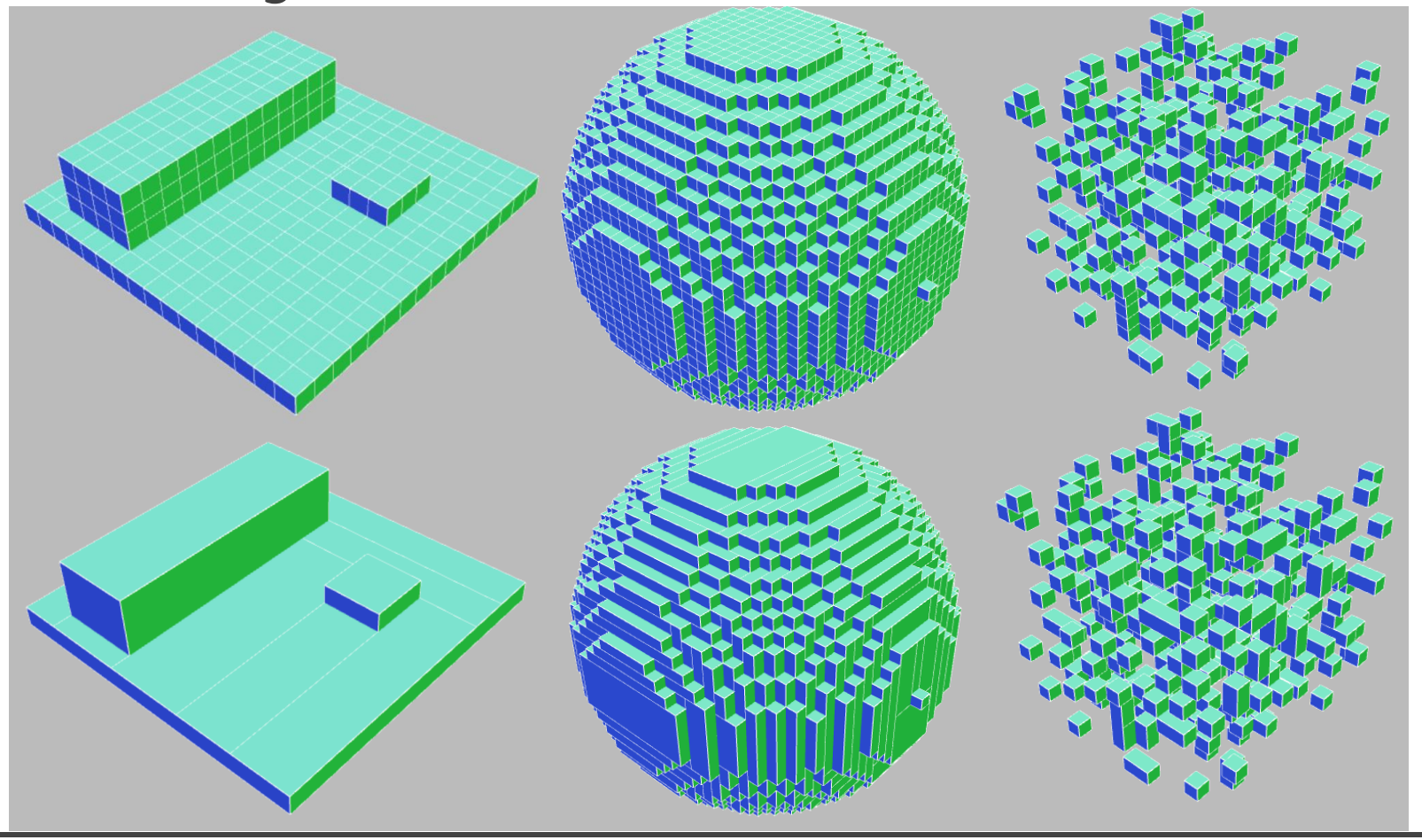
Stupid Method



Culling



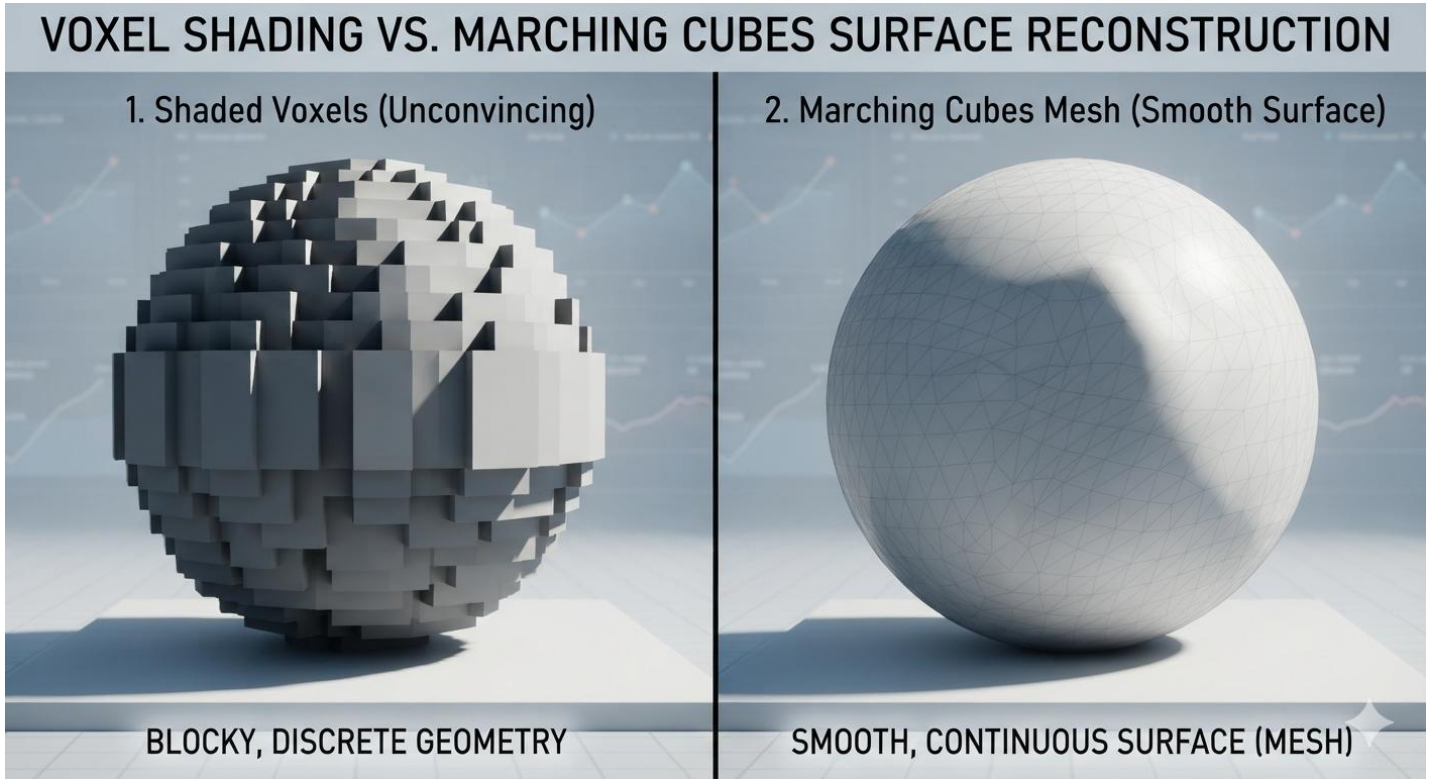
Greedy Meshing



자료 출처 : https://0fps.net/2012/06/30/meshing-in-a-minecraft-game/

Marching Cubes (1/2)

- "Marching cubes: A high resolution 3D surface construction" [William E. Lorensen and Harvey E. Cline / SIGGRAPH, 1987]



Voxel vs Marching Cubes made by NanoBanana

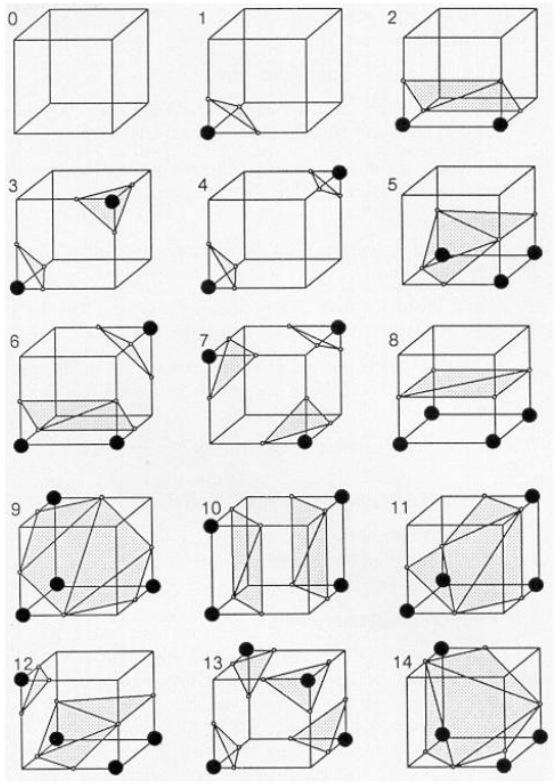
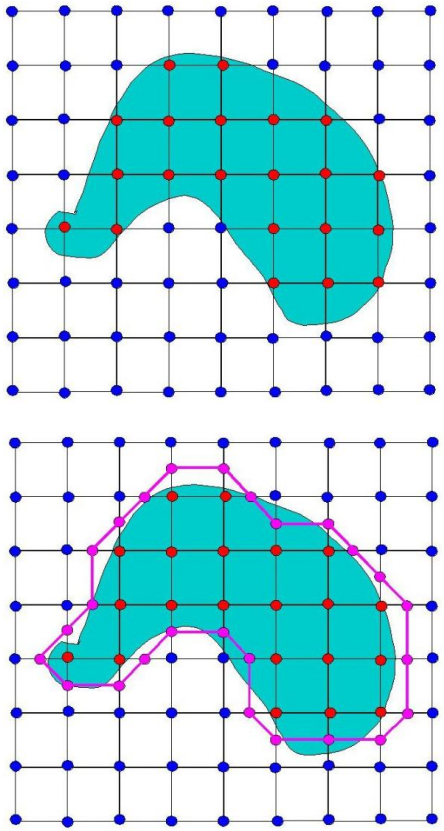


Figure 3. Triangulated Cubes.



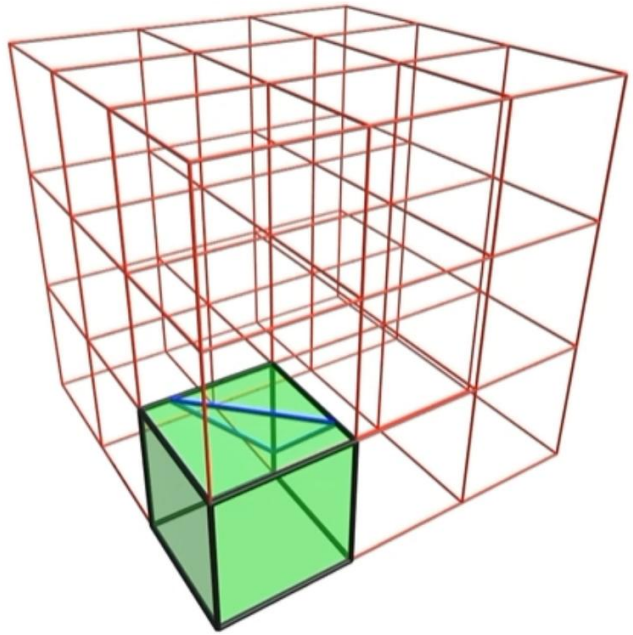
2D Marching Cube

자료 출처 : https://dl.acm.org/doi/epdf/10.1145/37402.37422 , https://www.cs.carleton.edu/cs_comps/0405/shape/marching_cubes.html

Marching Cubes (2/2)

- “Marching cubes: A high resolution 3D surface construction” [William E. Lorensen and Harvey E. Cline / SIGGRAPH, 1987]

Implementation

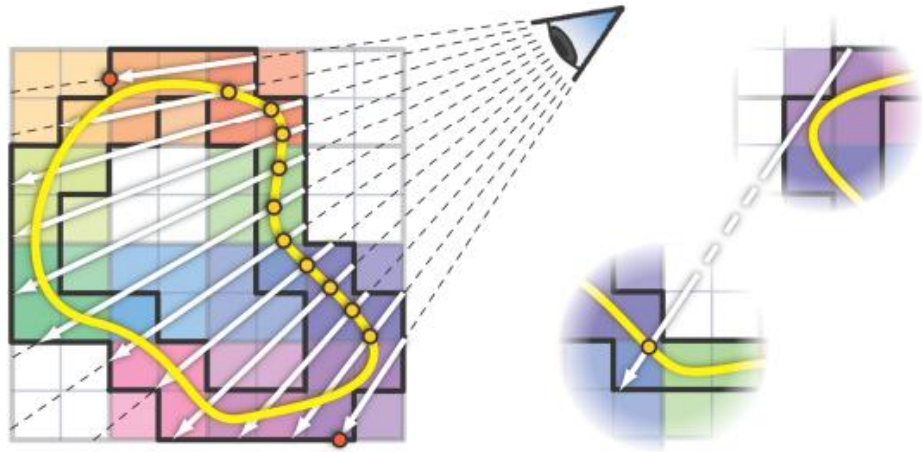


자료 출처 : https://youtu.be/B_xk71YopsA?si=3k6s4NCK1GeAVnXt

Level-Set Rendering (1/2)

- "Real-Time Ray-Casting and Advanced Shading of Discrete Isosurfaces" [Markus Hadwiger *et al* / Eurographics, 2005]

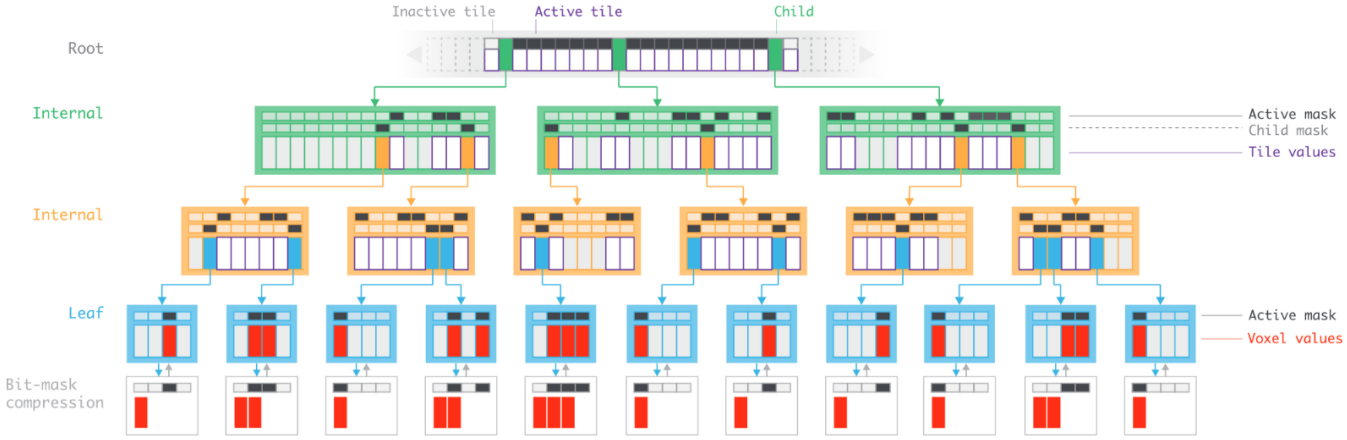


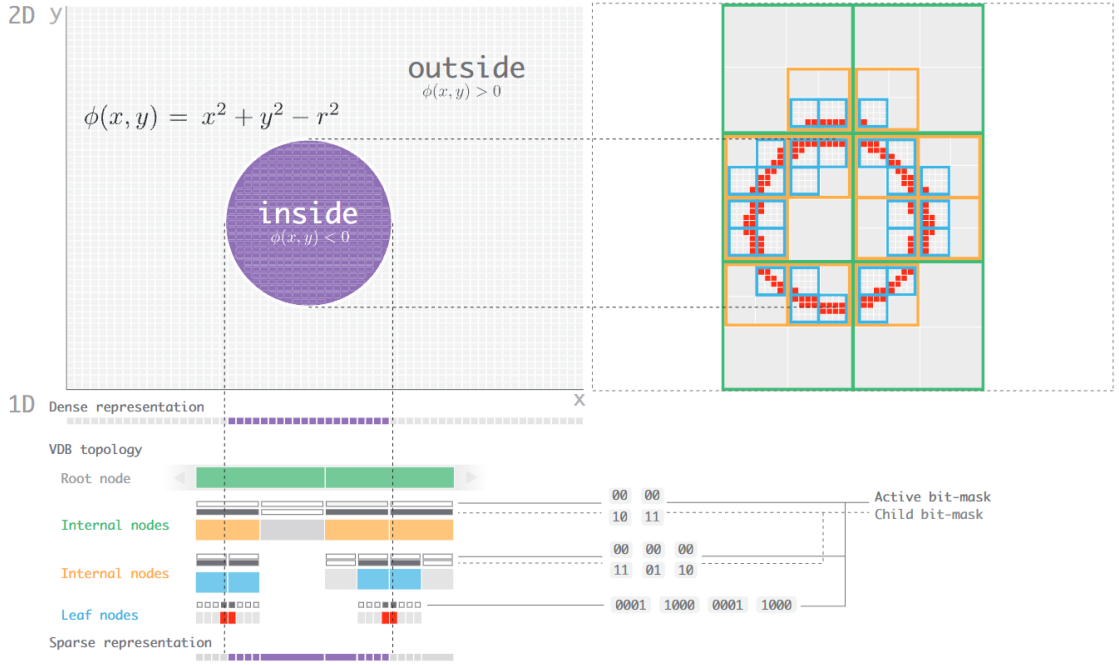


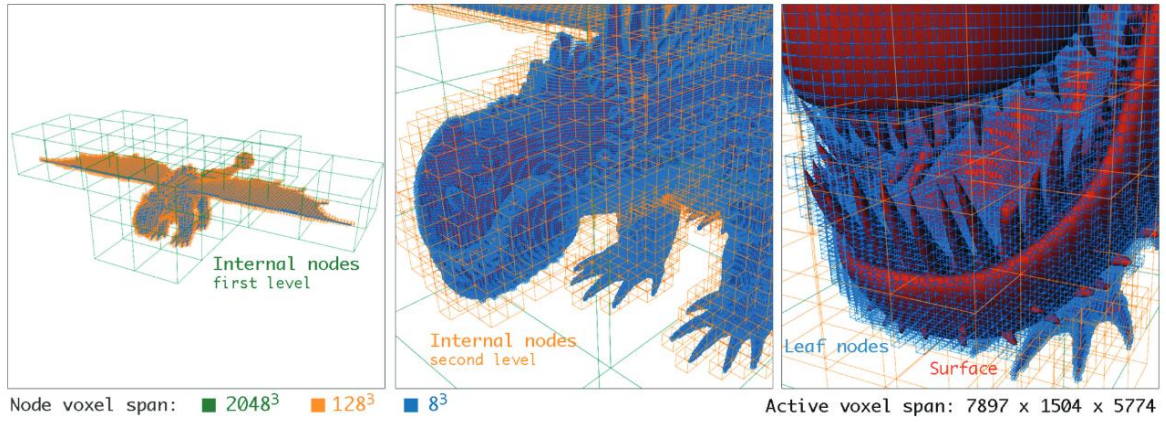
Voxel Rendering

Level-Set Rendering (2/2)

- "VDB: High-Resolution Sparse Volumes with Dynamic Topology" [Ken Museth / SIGGRAPH, 2013]







자료 출처 : https://www.museth.org/Ken/Publications_files/Museth_TOG13.pdf

OpenVDB Reel

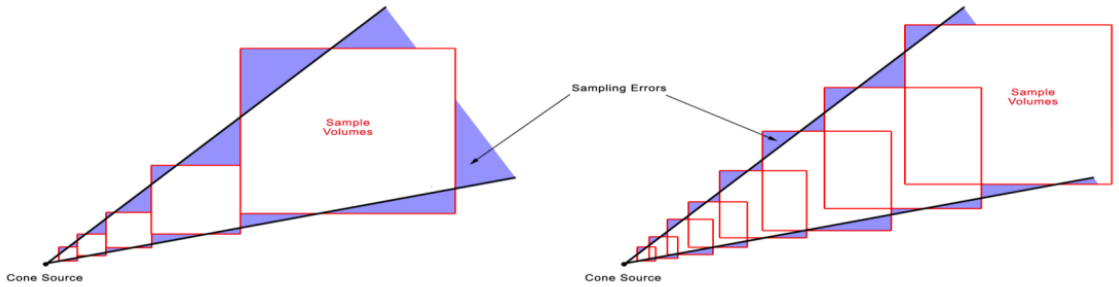


The standard C++ library and tools for sparse volumetric data used in visual effects and animation.

자료 출처 : https://youtu.be/GR2_8d9Bkc0?si=hzlVDAoDijXJ0u-F (OpenVDB Reel)

Voxel Rendering
Voxel Cone Tracing

- "Playing with Real-Time Shadows" [Nikolas Kasyan / SIGGRAPH, 2013]
  - Sampling Errors

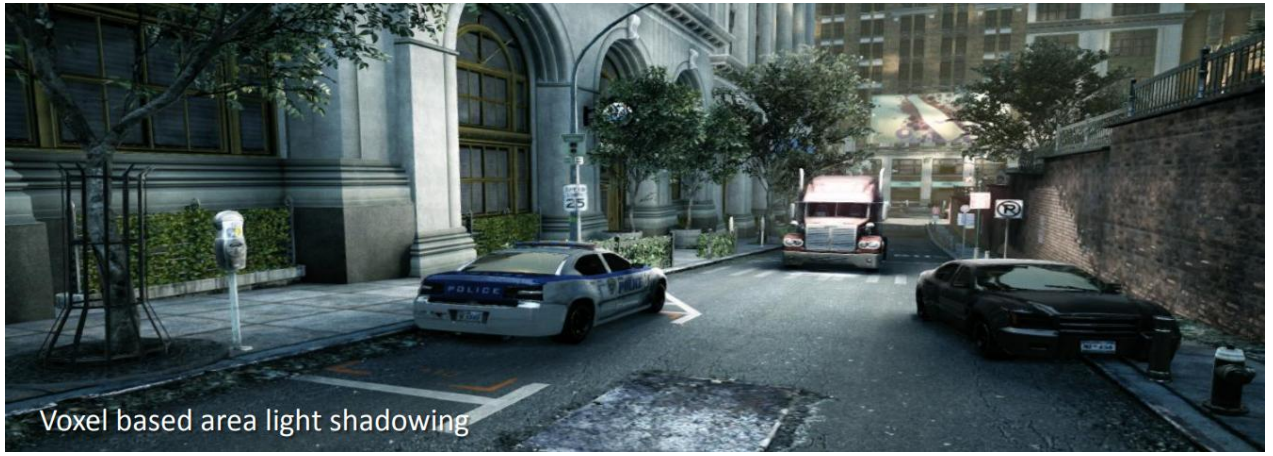




Ray-traced Spherical Area Light
Autodesk Maya 2012 ~40s

Real-time Area light shadows ~20ms
(Voxelization + Cone Tracing)

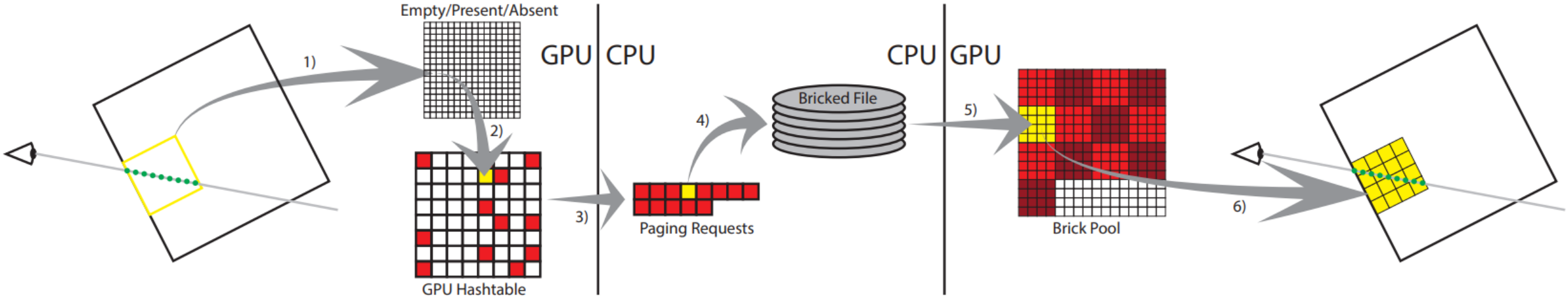




자료 출처 : https://www.realtimeshadows.com/sites/default/files/Playing%20with%20Real-Time%20Shadows_0.pdf

최근 GPU 구조 탐색 동향

- "An Analysis of Scalable GPU-Based Ray-Guided Volume Rendering" [Thomas Fogal *et al* / IEEE LDAV, 2013]

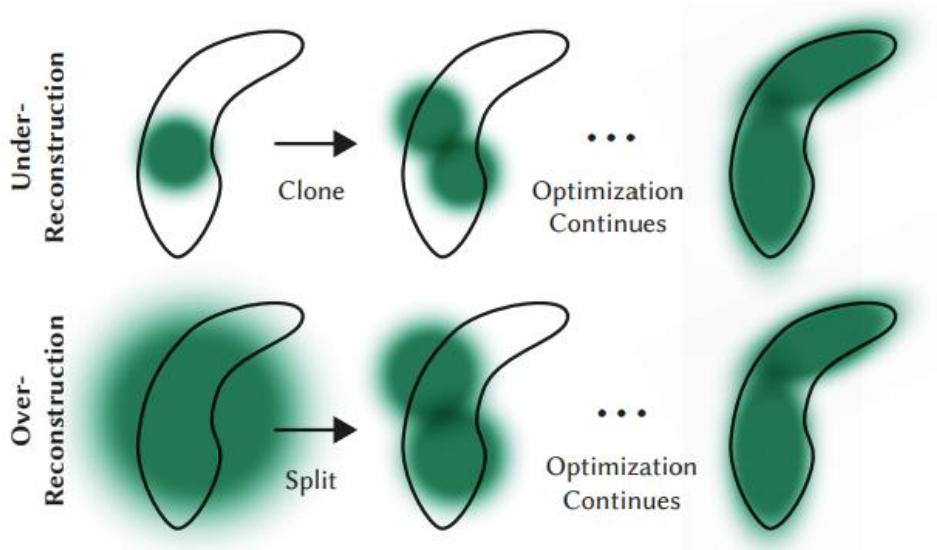


자료 출처 : https://cgvis.de/publications/2013/Fogal_2013_AAOS.pdf

채신 기술

- "3D Gaussian Splatting for Real-Time Radiance Field Rendering" [Bernhard Kerbl *et al* / SIGGRAPH, 2023]



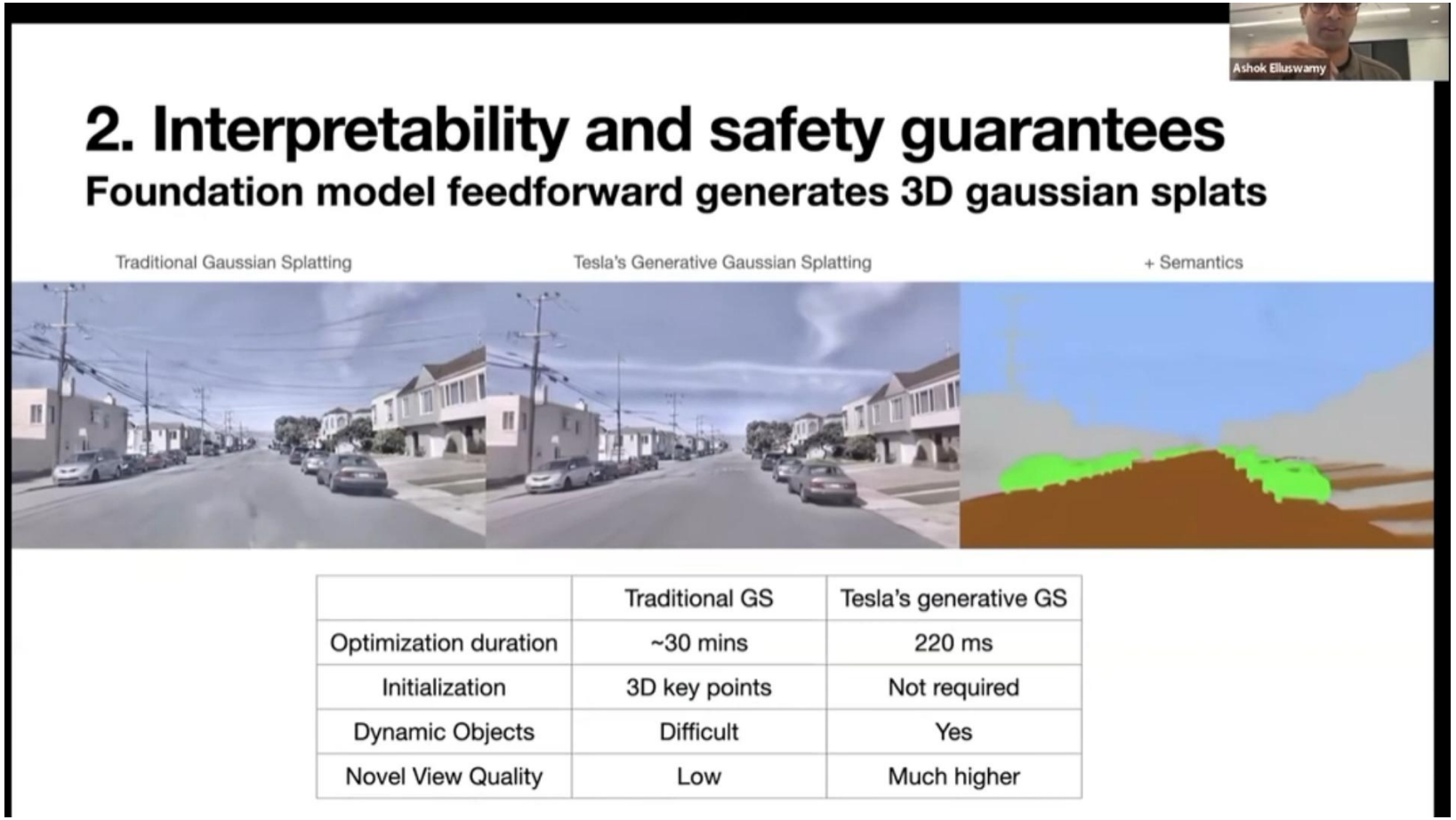


자료 출처 : https://youtu.be/T_kXY43VZnk?si=LF4818bO8EmXfTSt (3D Gaussian Splatting for Real-Time Radiance Field Rendering), https://cgvis.de/publications/2013/Fogal_2013_AAOS.pdf

기타 주제

채신 기술

- “Tesla’s Generative Gaussian Splatting”
  [Ashok Elluswamy / ICCV WDFM-AD Workshop, 2025]



자료 출처 : https://youtu.be/IRu-cPkpiFk?si=xMOp48TZCqjRnYME

Conclusion

- Beyond Polygons는 Triangle 중심 그래픽스를 확장
- 이미지·포인트·볼륨 기반 표현이 성능과 사실성을 향상
- GPU 발전과 함께 Hybrid Rendering이 핵심 트렌드
- 실시간 렌더링은 점점 더 데이터 중심으로 진화 중
- 미래 그래픽스는 Polygon과 Non-Polygon 기술의 융합 방향